50. Security Features

50.1 Features

- Armv8-M TrustZone security
  - Implementation Defined Attribution Unit (IDAU) is implemented
  - Security Attribution Unit (SAU) is implemented
    - 8 regions
  - Master Security Attribution Unit (MSAU) is implemented (IDAU for master other than CPU)
  - Code MRAM
    - Up to 2 regions (secure/non-secure)
  - SiP flash (Only for SiP product)
    - Up to 2 regions (secure/non-secure)
  - CM33 TCM
    - Up to 2 regions (secure/non-secure) for C-TCM and S-TCM
  - SRAM
    - Up to 2 regions (secure/non-secure) per SRAM
  - VBATT backup registers
    - Up to 2 regions (secure/non-secure)
  - Peripheral
    - Security attributes can be set individually for each unit/channel
  - CSC*1
    - CSC region is defined as non-secure
  - SDRAM*1
    - SDRAM region is defined as non-secure
  - OPSI0*1
    - OPSI0 region is defined as non-secure
  - OPSI1*1 (only for Standard product)
    - OPSI1 region is defined as non-secure
- Privileged control
  - Access permissions for memory except VBATT backup register are controlled by MPU managed by the privileged code
  - Privilege attributes of VBATT backup register are controlled by the registers of each controller
  - Individual privileged or Unprivileged attribution for each peripheral.
- Device lifecycle management
- Three debug levels
  - AL2: Non-secure and secure debug functions are enabled and accessible from the debugger
  - AL1: Only non-secure debug functions is enabled, and the debugger can access only defined non-secure debug accessible regions
  - AL0: No debug functions are available.
- Key injection
- Secure factory programming
  - Supports image programming in ciphertext format in an untrusted factory

- Secure boot
  - Support up to four revocable Root of Trust (RoT) in OTP, which is the SHA-256 hash digest of the RoT public key.
  - Verify the integrity and authenticity of image to be programmed
  - Verify the integrity and authenticity of the executable image by the immutable (OTP) First Stage Bootloader (FSBL) prior to executing the image.
- Cryptographic accelerator
  - See section 51, Renesas Secure IP (RSIP-E50D).
- Secure pin multiplexing
  - All I/O port pins can be configured individually as secure or non-secure
  - Peripheral pin function is valid when the security attribute of peripheral and I/O port match. See section 20, I/O Ports.
- Decryption on the fly
  - Decryption on the fly (DOTF) decrypts the encrypted content stored in the external memory in real time. See section 45, Decryption On The Fly (DOTF).

Note 1. There is no TrustZone filter in the external RAM and external device area. Therefore, even if these regions are marked as secure in the SAU settings, non-secure masters except the CPU can access them.

50.2 Tamper Detection

The device is provided with I/O ports capable of detecting an external tamper attempt. In the case of a tamper event:

- Timestamping — The RTC can take a timestamp to log the event, and an interrupt can be generated. See section 26, Realtime Clock (RTC).
- Hardware Unique Key — Hardware Unique Key (HUK) is zeroized (set to zero) following detection of a tamper event. See section 58, MRAM.
- VBATT backup registers — VBATT backup registers are zeroized (set to zero) following detection of a tamper event. See section 12, Battery Backup Function.

50.3 Arm Security Features

50.3.1 Arm TrustZone Technology

Arm TrustZone technology divides the system and the application into secure and non-secure domains. A secure application can issue both secure and non-secure transactions, but a non-secure application can only issue non-secure transactions. Secure transactions can only access secure memory and resources, and non-secure transactions can only access non-secure memory and resources. Secure transactions can be issued only using secure region addresses and non-secure transactions can be issued only using non-secure region addresses. Figure 50.1shows the security attributes of transactions that can be issued by each master.

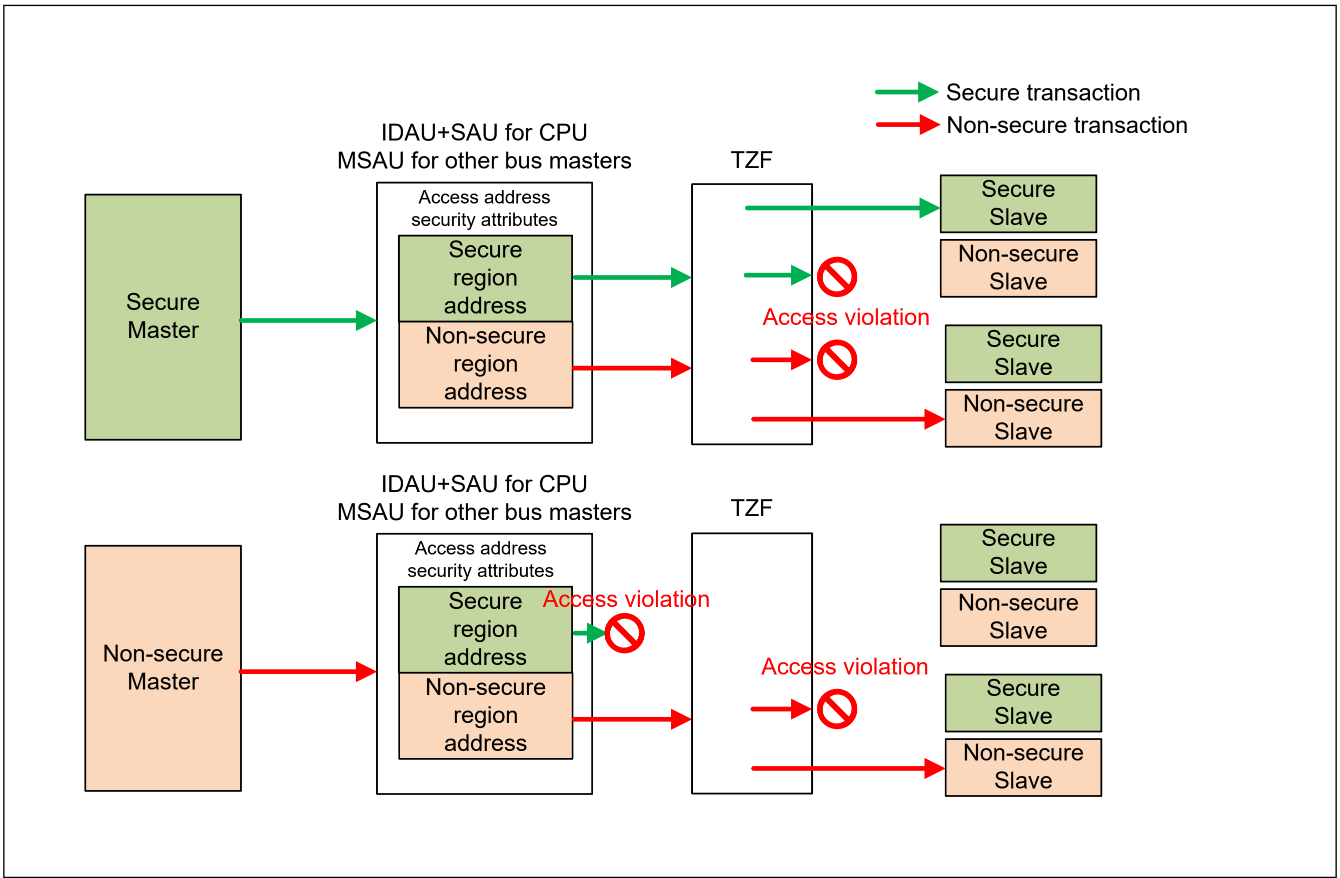


Figure 50.1 Transactions that can be issued by each master

For more details, see the *Arm® v8-M Architecture Reference Manual* and *Arm® Platform Security Architecture Trusted Base System Architecture for Arm®v6-M, Arm®v7-M and Arm®v8-M*.

50.3.2 Privileged Control

Systems and applications are divided into privileged and unprivileged domains. The CPU can limit or exclude access to some resources by executing code in privileged or unprivileged mode. Privileged mode can access both privileged and unprivileged domains, but unprivileged mode can access only the unprivileged domain.

For more details, see the *Arm® v8-M Architecture Reference Manual* and *Arm® Platform Security Architecture Trusted Base System Architecture for Arm®v6-M, Arm®v7-M and Arm®v8-M* .

50.3.3 Security Attribution

The TrustZone for Armv8-M implementation consists of the Security Attribution Unit (SAU) and Implementation Defined Attribution Unit (IDAU). The 4GB memory space is partitioned into Secure (S) and Non-Secure (NS) memory regions. The Secure memory space is further divided into two types, Non-secure Callable (NSC) and Secure.

Note:
- S = Secure addresses are used for memory and peripherals that are only accessible by secure software or secure masters.
- NSC = A special type of secure location. This type of memory is the only type in which an Armv8-M processor permits to hold a Secure Gateway (SG) instruction that enables software to transition from Non-secure to Secure state.
- NS = Non-secure addresses are used for memory and peripherals accessible by all software running on the device.

50.3.3.1 Implementation Defined Attribution Unit (IDAU)

The IDAU defines the code, SRAM and peripheral region into the secure alias region and the non-secure alias region by the address bit [28]. The secure code region and secure SRAM region are assigned the NSC security attributes. The security map defined by IDAU is fixed in hardware and cannot be changed by software.

50.3.3.2 Master Security Attribution Unit (MSAU)

MSAU is IDAU that defines system-specific security address map for masters other than CPU. The MSAU defines secure and non-secure alias regions but does not define the Non-secure Callable (NSC) and region number. Masters other than the CPU can issue a security transaction using the secure alias address defined by MSAU. However, non-secure masters are prohibited from issuing secure transactions using an address in the secure alias region. Figure 50.2 shows the defined security map of IDAU and MSAU. The security map defined by MSAU is fixed in hardware and cannot be changed by software.

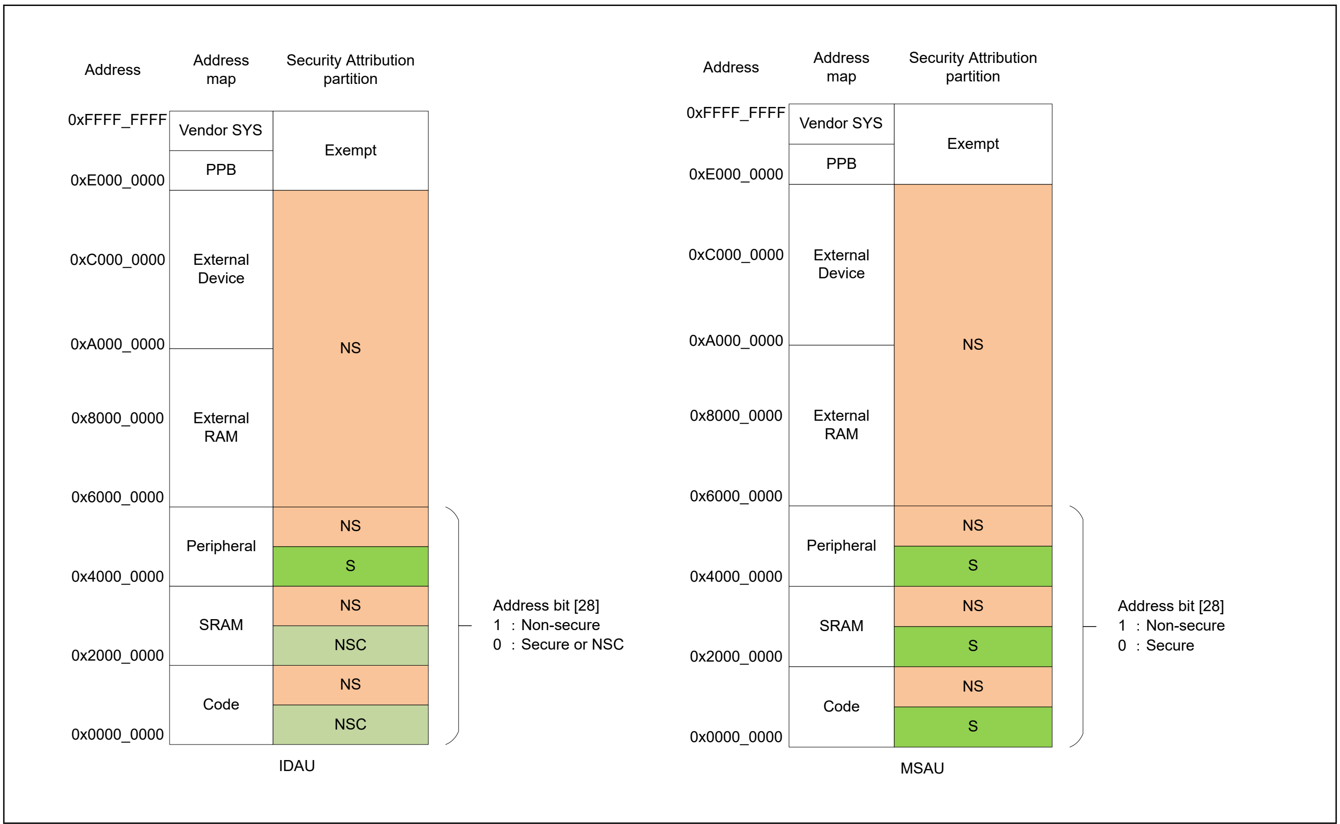


Figure 50.2 IDAU and MSAU defined security map

50.3.3.3 Secure Attribution Unit (SAU)

The SAU is a programmable unit that determines the security of an address. The SAU is programmable in the Secure state and has a programmers' model similar to the MPU. If an address maps to regions defined by both IDAU and SAU, the region of the highest security level is selected. Secure master can issue secure and non-secure transactions using the address of each security alias region. Non-secure master cannot issue secure transactions using the address of the secure alias region. Figure 50.3 shows the final determination of the address map security level.

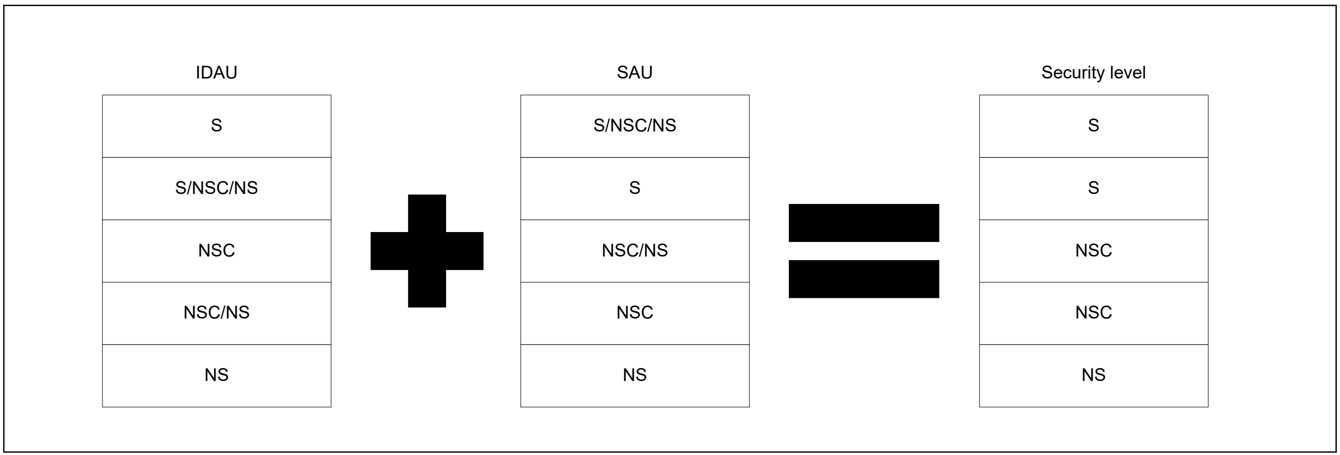


Figure 50.3 Final determination of address map security level

When using TrustZone to perform secure and non-secure region separation, SAU MUST be set according to the following.

The regions set as NS attribute in IDAU MUST be set to NS in the SAU as well. The regions set to NS attribute in IDAU are:

0x1000_0000 to 0x1FFF_FFFF (SAU Region 1 in Figure 50.4)

0x3000_0000 to 0x3FFF_FFFF (SAU Region 3 in Figure 50.4)

0x5000_0000 to 0xDFFF_FFFF (SAU Region 3 in Figure 50.4)

At least one NSC region MUST be created within any region defined as NSC by the IDAU. The regions set to NSC attribute by the IDAU are:

0x0000_0000 to 0x0FFF_FFFF (SAU Region 0 in Figure 50.4)

0x2000_0000 to 0x2FFF_FFFF (SAU Region 2 in Figure 50.4)

If you do not wish to define any isolation by means of using TrustZone, do not change SAU_CTRL.ALLNS = 0 and SAU_CTRL.ENABLE = 0 (initial value).

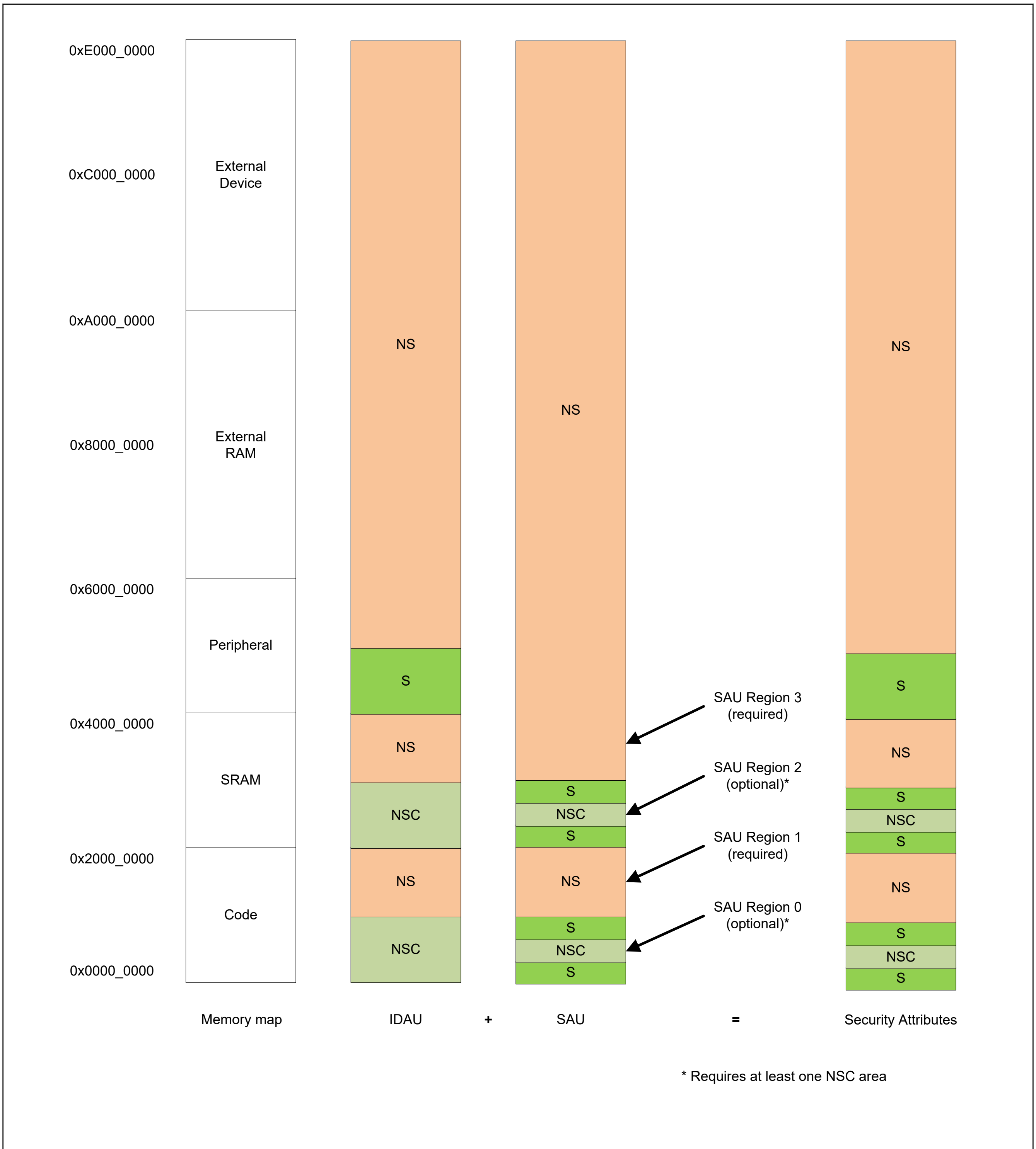


Figure 50.4 SAU settings and resulting security attributes

50.3.3.4 Region Number

The SAU and IDAU also define region numbers for each of the memory regions and security attributes. This region number is used by software to determine if a contiguous range of memory shares common security attributes. Figure 50.5 shows the defined region number of IDAU.

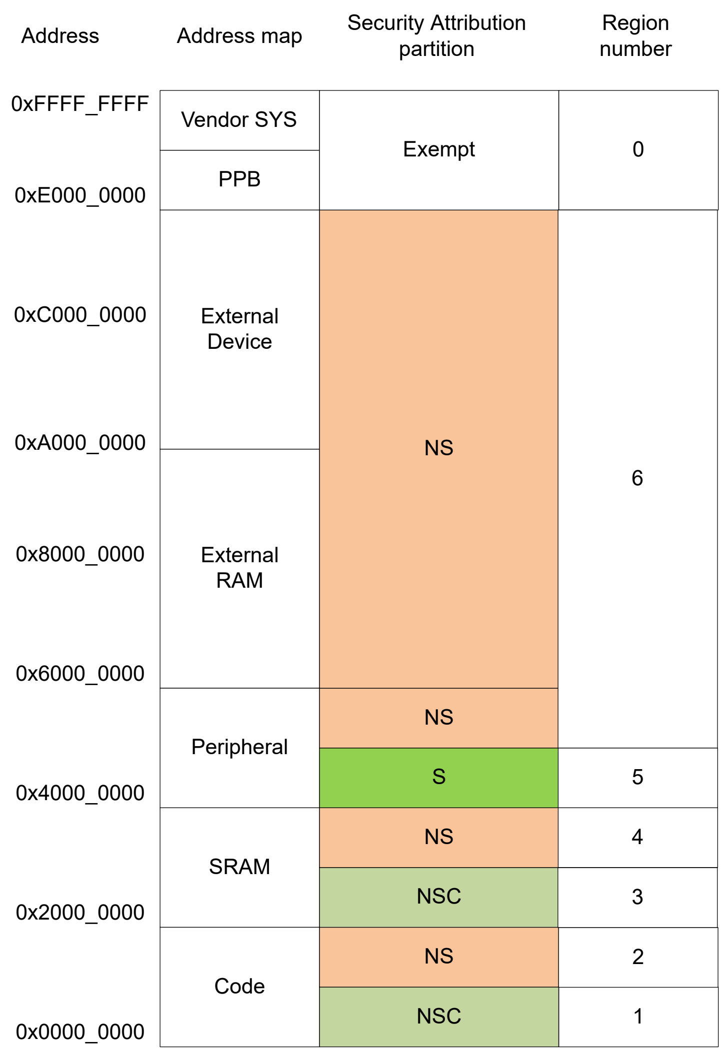


Figure 50.5 IDAU defined region numbers

50.3.3.5 Memory Security Attribution of TrustZone Filter

The memories are divided into S and NS regions. The memory security attribution of code MRAM and SiP flash (when equipped) is stored into nonvolatile memory by a boot firmware command when the device lifecycle is in the OEM state and the authentication level is AL2. These memory security attributions are applied before application execution. They cannot be updated by the application but are readable using CMSAMON or SFSAMON registers. The memory security attributions of CM33 TCM, SRAM and VBATT backup registers are set by a dedicated security attribution register writable only by secure access.

The code MRAM can be divided in up to two regions. The SiP flash can be divided in up to two regions. Programming or erasing of SiP flash should be managed by secure user to prevent non-secure user from programming or erasing the secure area. The CM33 TCM, SRAM and VBATT backup registers can be divided in up to two regions.. Figure 50.6 shows the memory mapping. Table 50.1 shows size of the memory region and Table 50.2 shows access permission of the memory region.

SRAM ECC region can be used as data region when ECC function is disabled. The security attribution of this region is set by the dedicated security attribute register. See section 57, SRAM for the details.

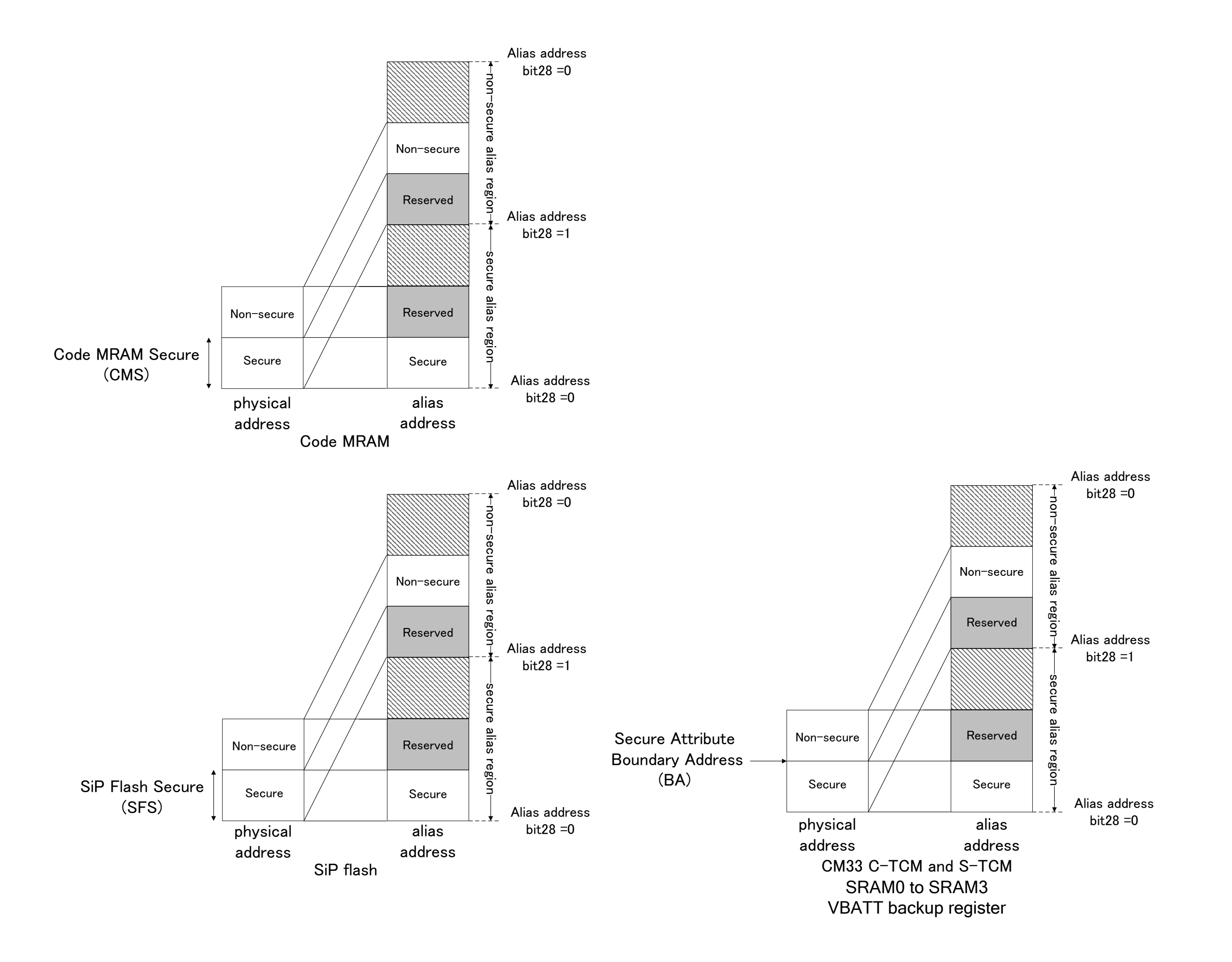


Figure 50.6 Memory mapping

Table 50.1 Size of memory region (1 of 2)

Memory region	Start address	Size
CM33 C-TCM secure	0x0000_0000*1	BA (8 KB)
CM33 C-TCM non-secure	0x1000_0000 + BA (8 KB)*1	C-TCM size - BA (8 KB)
Code MRAM secure	0x0200_0000	CMS size (32 KB)
Code MRAM non-secure	0x1200_0000 + CMS size (32 KB)	Code MRAM size - CMS size (32 KB)
SiP flash secure	0x0800_0000	SFS size (32 KB)
SiP flash non-secure	0x1800_0000 + SFS size (32 KB)	SiP flash size - SFS size (32 KB)
CM33 S-TCM secure	0x2000_0000*1	BA (8 KB)
CM33 S-TCM non-secure	0x3000_0000 + BA (8 KB)*1	S-TCM size - BA (8 KB)
SRAM0 secure	0x2200_0000	BA (8 KB)
SRAM0 non-secure	0x3200_0000 + BA (8 KB)	SRAM0 size - BA (8 KB)
SRAM1 secure	0x2208_0000	BA (8 KB)
SRAM1 non-secure	0x3208_0000 + BA (8 KB)	SRAM1 size - BA (8 KB)
SRAM2 secure	0x2210_0000	BA (8 KB)
SRAM2 non-secure	0x3210_0000 + BA (8 KB)	SRAM2 size - BA (8 KB)
SRAM3 secure	0x2218 0000	BA (8 KB)

Table 50.1 Size of memory region (2 of 2)

Memory region	Start address	Size
SRAM3 non-secure	0x3218 0000 + BA (8 KB)	SRAM3 size - BA (8KB)
VBATT backup register secure	0x4001_ED00	BA (32 bytes)
VBATT backup register non-secure	0x5001_ED00 + BA (32 bytes)	Backup register size - BA (32 bytes)

Note: The number in parentheses indicates the smallest unit that can be set by the user.
Note: BA is the setting value of the security attribution boundary address register for each memory region.
Note 1. Address of CPU1. For address of other bus masters, see Table 2.37.

Table 50.2 Access permission of the memory region

Memory region	Secure transaction	Non-secure transaction
Each memory region configured as S or NSC	Allowed	Write ignored/Read ignored TrustZone access error is generated
Each memory region configured as NS	Write ignored/Read ignored TrustZone access error is generated	Allowed

50.3.3.6 CPU Security Attribution

Cortex-M33 security extension can be disabled by the boot firmware command only when cortex-M33 is configured as the secondary CPU in dual core product. Disabling the Cortex-M33 security extension is a permanent setting.

50.3.3.7 Peripheral Security and Privilege Attribution of TrustZone Filter

Each peripheral can be configured as secure or non-secure and privileged or unprivileged. Peripherals are divided into two types.

Type1 peripheral has one security and privilege attribution and access to all registers is controlled by one security and privilege attribution. Type1 peripheral security and privilege attribution is set to the PSARx (x = B to E) and the PPARx (x = B to E) registers by the secure and privileged application.

Type2 peripheral has the security and the privilege attribution for each register or for each bit and access to each register or bit field is controlled according to these security and privilege attributions. Type2 peripheral security and privilege attribution is set to the Security Attribution register and the Privilege Attribution register in each module by the secure and privileged application. For details on the Security Attribution register and the Privilege Attribution register, see each section.

Table 50.3 shows the classification of peripheral type.

Table 50.3 Classification of peripheral type

Type1	Type2
SCI, SPI, OSPI, DOTF, USBFS, IIC, I3C, RSIP-E50D, ESWM, ESC, CANFD, DOC, SDHI, CRC, CAC, ACMPHS, TSN, ADC16H, DAC12, POEG, DSMIF, PDG, AGT, GPT, ULPT, RTC, IWDT, and WDT	System control (Resets, PVD, Clock Generation Circuit, Low Power Modes, Battery Backup Function, Register Write Protection Function), MRAM controller, SRAM controller, CPU cache, DMAC, DTC, ICU, MPU, BUS, Security setting, ELC, and I/O ports

Table 50.4 shows the access permission of type1 peripherals. The access permission of type2 peripherals is different by peripherals. Each register description of type2 peripherals includes the note of S-TYPE and P-TYPE indicating the specification of the access permission. For details on S-TYPE and P-TYPE, see section Appendix 4, Notes for Register R/W.

Table 50.4 Access permission of type1 peripherals (1 of 2)

	Secure and Privileged access	Secure and Unprivileged access	Non-secure and Privileged access	Non-secure and Unprivileged access
Peripheral configured as Secure and Privileged	Allowed	Write ignored/Read ignored TrustZone Access error is generated	Write ignored/Read ignored TrustZone Access error is generated	Write ignored/Read ignored TrustZone Access error is generated
Peripheral configured as Secure and Unprivileged	Allowed	Allowed	Write ignored/Read ignored TrustZone Access error is generated	Write ignored/Read ignored TrustZone Access error is generated

Table 50.4 Access permission of type1 peripherals (2 of 2)

	Secure and Privileged access	Secure and Unprivileged access	Non-secure and Privileged access	Non-secure and Unprivileged access
Peripheral configured as Non-secure and Privileged	Write ignored/Read ignored TrustZone Access error is generated	Write ignored/Read ignored TrustZone Access error is generated	Allowed	Write ignored/Read ignored TrustZone Access error is generated
Peripheral configured as Non-secure and Unprivileged	Write ignored/Read ignored TrustZone Access error is generated	Write ignored/Read ignored TrustZone Access error is generated	Allowed	Allowed

50.3.4 TrustZone Access Error

Table 50.5 shows the behavior when TrustZone access error occurs. The behavior varies depending on the master.

Table 50.5 Behavior of TrustZone access error

DAP	CPU	DMAC/DTC	ESWM
• Only error response is returned*1	• IDAU/SAU detects SecureFault exception • TrustZone filter detects BusFault exception • Can issue IRQ or reset*2	• Stop transfer • Issue IRQ or reset*2 • Generate interrupt (DMAn_TRANSERR)	• Issue IRQ or reset*2 • Generate interrupt (ETHER_GWEI)*3

Note: This behavior is not applicable for bufferable write access. For more information on when a bufferable write access error is detected, see section 15, Buses.
Note 1. When a TrustZone access error occurs by a debugger access, exception, NMI, or reset does not occur. Only the error response is returned.
Note 2. The operation after error detection is selected by the BUSOAD.SLERROAD bit.
Note 3. AXI error status in GWEIS0.AES bit is set. The interrupt occurs when interrupt is enabled in GWEIE0.AEE bit.

50.4 Device Lifecycle Management

Device Lifecycle Management (DLM) identifies the current development/production/deployment phase of the device and controls the capabilities of the debug function, the serial programming interface, and Renesas test mode. Figure 50.7 shows the available device lifecycle states and Table 50.6 shows the lifecycle state definitions and capabilities in each state.

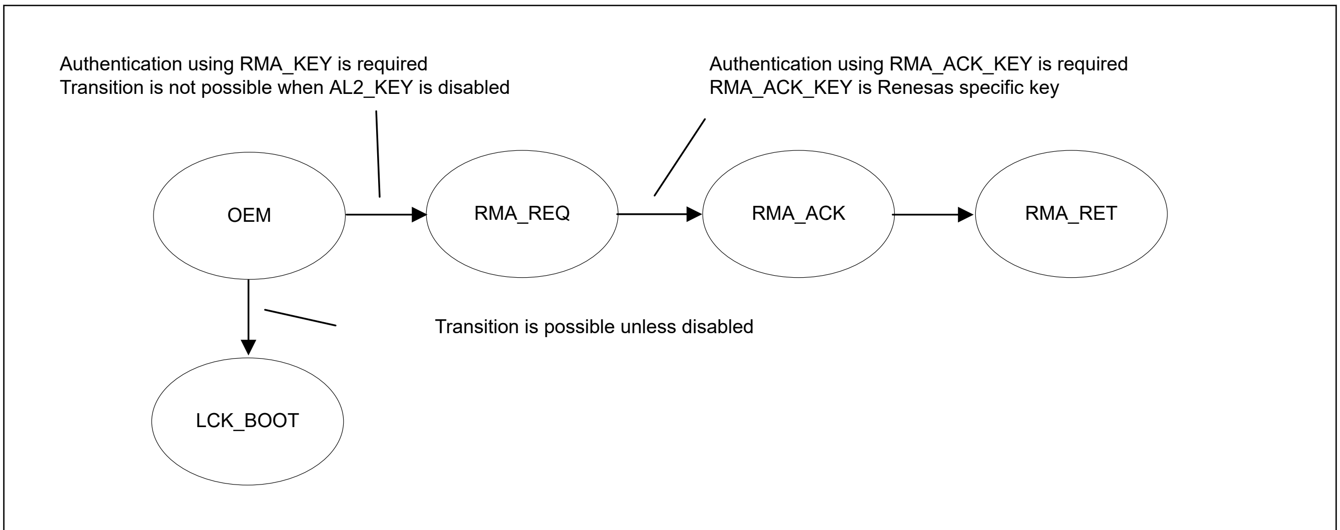


Figure 50.7 Device lifecycle states

Table 50.6 Lifecycle state definition and capabilities in each state (1 of 2)

Lifecycle	Definition	Protection level	Debug function	Serial programming	Renesas test mode
OEM	"Original Equipment Manufacturer" The device is owned by the customer.	PL2 or PL1 or PL0	Depend on the authentication level		Not available

Table 50.6 Lifecycle state definition and capabilities in each state (2 of 2)

Lifecycle	Definition	Protection level	Debug function	Serial programming	Renesas test mode
LCK_BOOT	"LoCKed BOOT interface" The debug interface and the serial programming interface are permanently disabled.	PL0	Not available	Not available	Not available
RMA_REQ	"Return Material Authorization REQuest" Request for RMA. The customer must send the device to Renesas in this state.	PL0	Not available	Available Cannot access code MRAM/SiP flash area	Not available
RMA_ACK	"Return Material Authorization ACKnowledged" Failure analysis in Renesas	PL2	Available in the secure and non-secure debug	Available Cannot access code MRAM/SiP flash area	Available
RMA_RET	"Return Material Authorization RETurn" The device is back to the customer. The device does not boot.	PL0	Not available	Not available	Not available

50.4.1 Changing the Lifecycle State

Use the boot firmware commands to change the device lifecycle state. See the boot firmware application note for details of the command. The lifecycle state cannot be updated by an application, but the current lifecycle state can be read through the DLMMON register.

As shown in Figure 50.7, each transition is one-way and the state cannot be regressed.

Transition from OEM to RMA_REQ requires key authentication using the RMA_KEY which must have been previously injected by the customer. The key length of the RMA_KEY is 128 bits. Inject the RMA_KEY as shown in section 50.5. Secure Key Injection. RMA_KEY can be injected in AL2.

The key authentication uses a challenge and response authentication or authentication using the MCU's unique ID. The response (challenge and response authentication) or the authentication code (using the MCU's unique ID) can be calculated as follows:

Response = AES-128 CMAC (RMA_KEY, 128-bit challenge)

Authentication code = AES-128 CMAC (RMA_KEY, 128-bit MCU unique ID)

The contents of the MRAM and SiP flash except permanently locked blocks or registers are erased when transitioning to RMA_REQ. The contents of the permanently locked blocks or registers can be read by Renesas at failure analysis. An MRAM block can be permanently locked by setting the PBPS/PBPS_SEC registers to permanently disable writing of the block. The option-setting memory can be permanently locked by the POFSPS register, permanently disabled writing of the register. Transition to RMA_REQ is not possible if the AL2_KEY is disabled. The MCU does not respond after changing the device lifecycle state to RMA_REQ. To continue to use boot firmware commands, you must enter boot mode again after a reset. See the *Boot Firmware Application Note* for details.

Transition from OEM to LCK_BOOT is possible unless that transition has been explicitly disabled. Use the parameter setting command in AL2 or AL1 to prohibit the transition to LCK_BOOT. The LCK_BOOT transition prohibition is a permanent setting and cannot be undone. The debug interface and serial programming interface are permanently disabled in LCK_BOOT.

50.4.2 Protection and Authentication Level

The protection level (PL) and the authentication level (AL) determine the availability of the debug function and the serial programming interface. PL and AL are fixed except in the OEM state. For PL and for AL, three levels can be set for each in the OEM state. AL indicates a temporary authentication status and is initialized to PL after an MCU power-on reset. Both PL and AL can be changed only by the boot firmware and cannot be changed by the application. Figure 50.8 shows the available Protection Level and Authentication Level states and transitions. Table 50.7 shows the availability of the debug function and the serial programming interface in each level.

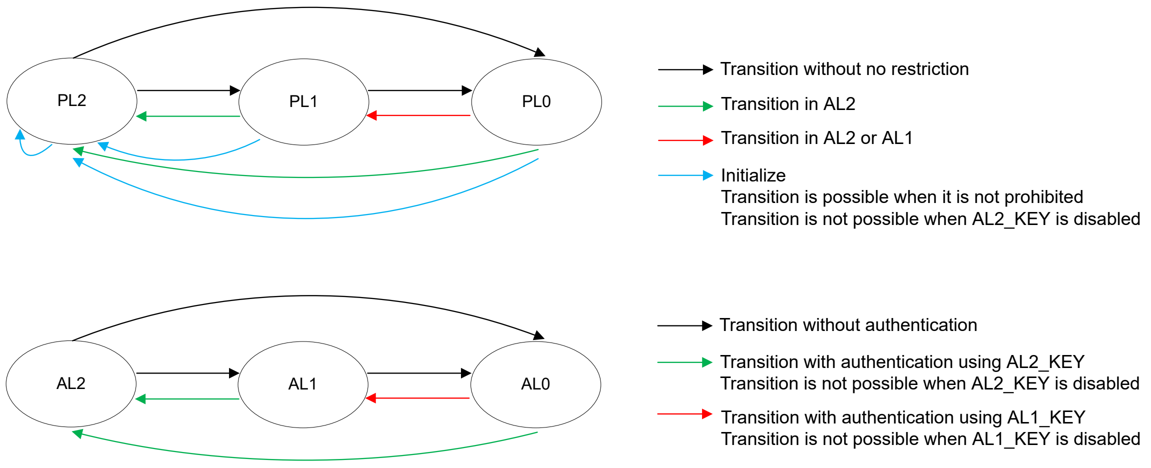


Figure 50.8 PL and AL states and transitions

Table 50.7 Availability of the debug function and the serial programming interface in each authentication level

AL	Debug function	Serial programming interface
AL2	Non-secure and Secure debug functions are enabled and accessible from the debugger	All functions are available
AL1	Only non-secure debug function is enabled, and debugger can access only defined non-secure debug accessible regions	Available but cannot program, erase, or read secure code MRAM or SiP flash area
AL0	No debug functions are available	Available but cannot access code MRAM or SiP flash area

Use the boot firmware commands to change PL and AL. See the boot firmware application note for details of the command.

Changing to a lower PL can be done with no restriction. Changing to a higher PL requires that the MCU be at that AL. For example, when the current AL is AL1, changing to PL1 is possible but changing to PL2 is impossible. PL can be reset by the Initialize command unless the command itself is disabled. After the Initialize command is executed, PL is set back to PL2 and the contents on the MRAM and SiP flash memory are erased. Initialize command does not execute when OTP area of option-setting memory is not blank. Initialize command can be disabled permanently in all AL states by parameter setting command to prevent memory erasure without authorization. Initialize command is disabled also when AL2_KEY is disabled. MCU does not respond after executing the Initialize command. To continue to use the serial programming commands, reenter the boot mode after a reset. See the *Boot Firmware Application Note* for the detail.

Changing to a lower AL can be done without authentication. Changing to a higher AL requires key authentication using AL2_KEY or AL1_KEY, as appropriate. These keys are 128-bit keys. Injection of AL2_KEY or AL1_KEY is performed as shown in section 50.5. Secure Key Injection. AL2_KEY can be injected in AL2, AL1_KEY can be injected in AL2 or AL1. The key authentication uses a challenge and response authentication. The response can be calculated as follows:

Response = AES-128 CMAC (KEY, 128 bits challenge)

AL2_KEY can be disabled permanently in AL2 with the parameter setting command.

AL1_KEY can be disabled permanently in AL2 or AL1 by parameter setting command.

50.4.3 Serial Programming

Whether a serial programmer can be connected and the range of memory that can be accessed depends on the device lifecycle state and AL as shown in Table 50.6 and Table 50.7. In addition, the accepted serial programming commands differ depending on the device lifecycle state and AL. See the *Boot Firmware Application Note* for details of the commands.

50.4.4 Device Lifecycle State and PL Change Example

The following section describes a typical device lifecycle state and PL change example.

- Secure developer
  - Set the memory security attribution of the code MRAM and SiP flash using the boot firmware command
  - Program and debug the secure application
  - If the AL2_KEY and RMA_KEY are required, inject them using the boot firmware commands
  - Inject application AES, ChaCha20-Poly1305, RSA, ECC, HMAC and EdDSA keys listed in Table 50.8 (if required)
  - Prepare the MCU for the non-secure developer
    - If the non-secure developer is not permitted to use the Initialize command, disable it using the boot firmware command.
    - If the AL2_KEY is not used and the non-secure developer is not permitted to use the Initialize function, disable the AL2_KEY using the boot firmware command.
    - Change the PL from PL2 to PL1 using the boot firmware command
- Non-secure developer
  - Program and debug the non-secure application.
  - If the AL1_KEY is required, inject it using the boot firmware command.
  - Inject application AES, ChaCha20-Poly1305, RSA, ECC, HMAC and EdDSA keys listed in Table 50.8 (if required)
  - Prepare the MCU for end-product deployment
    - Disable the Initialize command using the boot firmware command (if required)
    - If the AL1_KEY is not used, disable the AL1_KEY using the boot firmware command
    - Change the PL from PL1 to PL0 using the boot firmware command

50.4.5 Failure Analysis

If the customer requests failure analysis by Renesas, it is necessary to send the device after changing the device lifecycle state to RMA_REQ. If the device lifecycle state is not RMA_REQ, Renesas cannot perform the failure analysis. After failure analysis, Renesas changes the lifecycle to RMA_RET and returns the device to the customer.

Note: RMA_KEY is required to change the device lifecycle state to RMA_REQ or the MCU's Unique ID must be used as part of an authentication code. See section 50.4.1. Changing the Lifecycle State for details.

50.5 Secure Key Injection

To inject a user key into the MCU, perform the following steps.

Renesas provides the Security Key Management Tool, available on the Renesas website, to assist with key injection preparation.

1. Create a 256-bit installation key.
   This key is called the User Factory Programming Key (UFPK) and is used to wrap a user key
2. Get the wrapped version (W-UFPK) of the UFPK through the Renesas Key Wrapping Service.
3. Wrap the user key using the UFPK.
4. Send the W-UFPK and the wrapped user key to the MCU using a boot firmware command.
   The user key is unwrapped, wrapped with the MCU's hardware unique key, and stored in nonvolatile memory. DLM keys are stored in unmapped flash. Application keys are stored at the address specified with the key injection command.

Figure 50.9 shows an example of key injection and Table 50.8 shows the keys that can be injected.

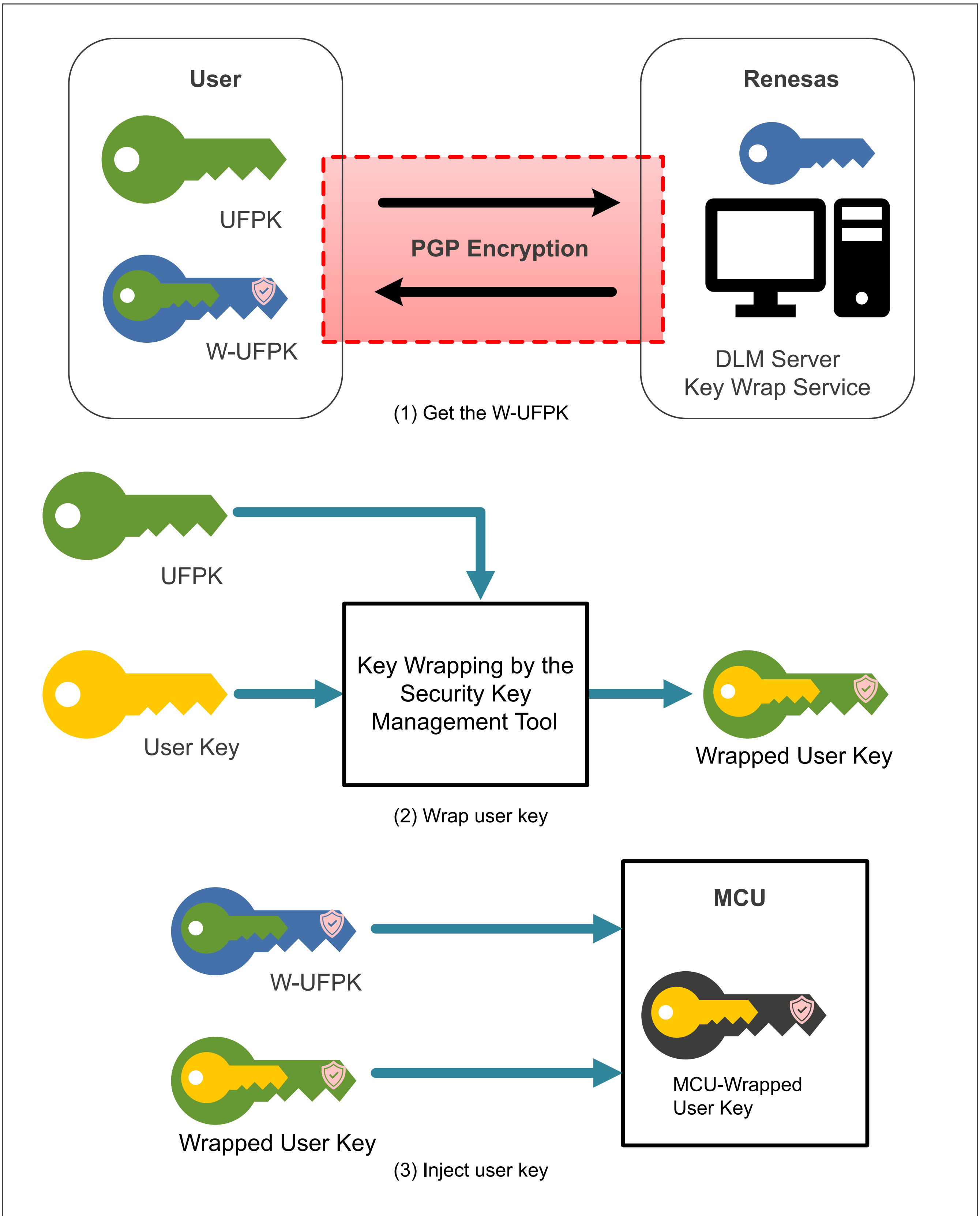


Figure 50.9 Key injection

Table 50.8 Keys that can be injected

DLM transition	AL transition	AES	ChaCha20-Poly-1305	RSA	ECC	HMAC	EdDSA	Secure boot	Other
RMA_KEY	AL2_KEY, AL1_KEY	AES-128, AES-192, AES-256, AES-XTS-128, AES-XTS-256	ChaCha20-Poly-1305	RSA-1024, RSA-2048, RSA-3072, RSA-4096 (Public and Private) RSA-2048 Public Key for TLS	ECC P-192, ECC P-224, ECC P-256, ECC P-384, ECC P-521 (Public and Private) ECC P256r1, ECC P384r1, ECC P512r1, ECC secp256k1 (Public and Private)	HMAC-SHA224, HMAC-SHA256, HMAC-SHA384, HMAC-SHA512, HMAC-SHA512/224, HMAC-SHA512/256, HMAC-SHA3-224, HMAC-SHA3-256, HMAC-SHA3-384, HMAC-SHA3-512	Ed25519 (Public and Private)	Up to 4 OEM_ROOT_PKs	Key-Update Key

50.6 Secure Factory Programming

In addition to secure key injection support for injecting user keys (DLM, debug authentication, application, and secure boot keys), the MCU supports the programming of a firmware image in ciphertext format to prevent assets from being leaked during production programming. This enables secure factory programming in a non-secure environment. Secure factory programming is supported by the boot firmware. The maximum capacity that can be programmed with the secure factory programming is almost 16 MB. See the security key management tool user's manual or the boot firmware application note for details of the capacity limitation. Figure 50.10 shows an example of secure factory programming of an encrypted firmware image. The customer wraps the Image Encryption Key with the UFPK and encrypts the image with the Image Encryption Key using AES128-CCM. When the customer sends the W-UFPK, the wrapped Image Encryption Key, and encrypted image to the MCU through the serial programming interface, the MCU decrypts and programs the firmware image.

To further support secure factory programming in a non-secure environment, the DLM state, Protection Level, and Authentication Keys can all be set via a single boot firmware command. Points to note about this boot firmware command include:

- Encrypted firmware programming can be performed only when the MCU is in the OEM state.
- This command changes the Protection Level of the MCU. The initial PL must be PL2. The final PL must be PL0.
- If the DLM state remains in the OEM state, AL2 key must be injected. AL1 key can optionally also be injected.
- The MCU can be transitioned to the LCK_BOOT state. In this case, AL keys cannot be injected.
- AL keys must be wrapped with the same UFPK as the Image Encryption Key.
- This command erases all code MRAM area except option-setting memory before programming encrypted firmware image. If there are any permanently locked blocks, this command cannot be executed.
- Prohibit to execute this command if the current register setting or write value to the registers related to startup area selection are other than the following:
  - SAS.BTFLG = 1b
- All option-setting memory values must be included in the encrypted firmware image, including those not changed from their default values. However, if the following areas are write-protected, the write data in these areas should not be included in the image. If included, this command terminates with an error.
  - Configuration setting area protected by POFSPS register.

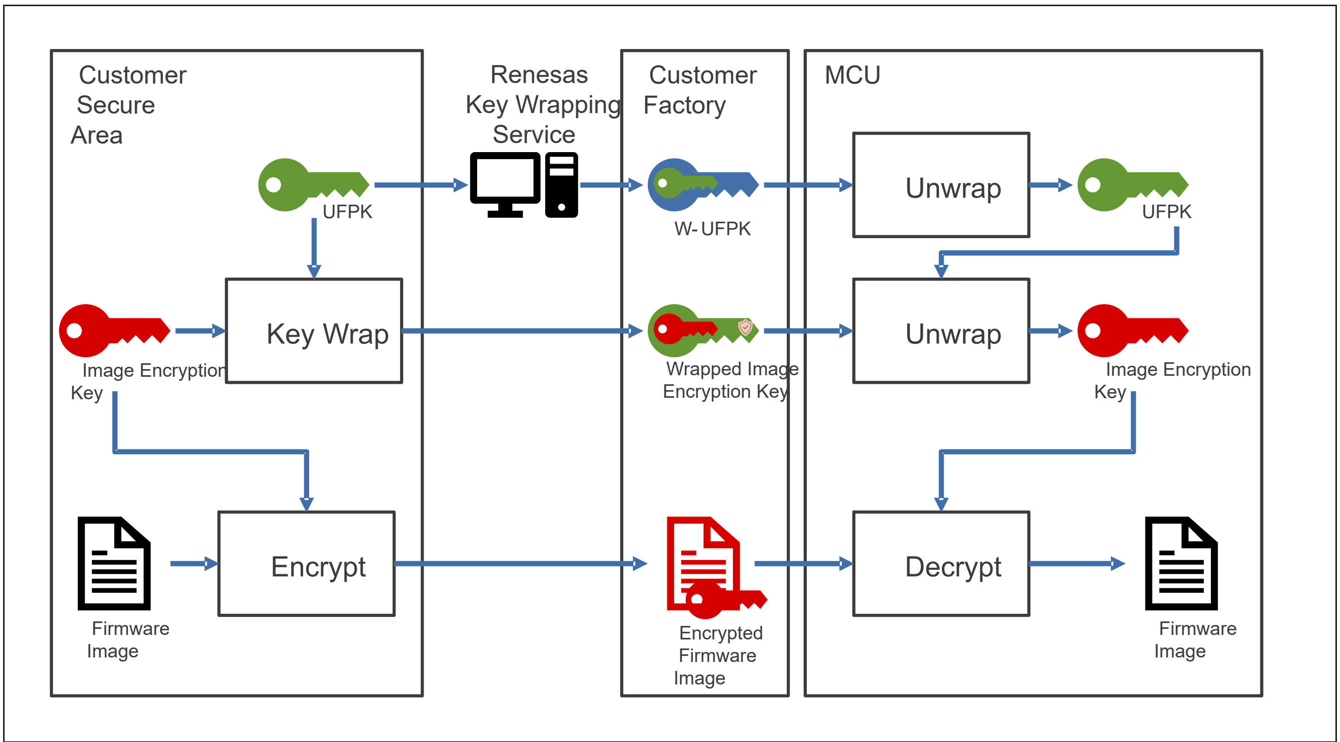


Figure 50.10 Secure factory encrypted image programming

50.7 Secure Boot

Secure boot verifies the integrity and authenticity of an OEM boot loader (called OEM_BL) starting at the beginning of application executable memory.

The OEM_BL is verified when it is initially programmed and prior to execution.

OEM_BL is verified when it is programmed using the boot firmware. Two certificates are used for validation. One is the key certificate, which is used to certify the OEM_BL verification key (called the OEM_BL_PK, the public part of the OEM_BL verification key pair). The key certificate is signed by the secret part of OEM_BL verification root key pair (called OEM_ROOT_SK). The public part of OEM_BL verification root key pair (called OEM_ROOT_PK) is registered with the MCU as the root of trust key using secure key injection as described in section 50.5. Secure Key Injection. The OEM_ROOT_SK and OEM_ROOT_PK are a 256-bit ECC (secp256r1 curve) key pair. The SHA2-256 hash of the OEM_ROOT_PK is stored during the secure key injection process. Up to four OEM_ROOT_PKs can be registered. These OEM_ROOT_PKs can be revoked if they are no longer needed. The validation process succeeds if the non-revoked first OEM_ROOT_PK with the biggest n value matches.

The other certificate for validation during programming is the code certificate, which is used to certify the OEM_BL. The code certificate is signed by the secret part of OEM_BL verification key pair (called OEM_BL_SK). After successful validation of the OEM_BL, the HMAC value of OEM_BL and the code certificate (called OEM_BL_digest) is generated and programmed to MRAM. HMAC SHA2-256 is used to generate the HMAC value. A derived key from the HUK is used as the HMAC key, so the HMAC value is unique in each MCU.

In single-chip mode, the immutable first stage boot loader (called FSBL) in the on-chip OTP is executed after reset when the FSBL is enabled in the FSBLCTRL0 register. When secure boot is selected in FSBLCTRL1 register, the FSBL generates the HMAC value of the OEM_BL and the code certificate, and compares it to the expected HMAC value. When the HMAC value is the same, the FSBL jumps to the OEM_BL. When CRC boot is selected in the FSBLCTRL1 register, the FSBL calculates the CRC of the OEM_BL and compares it to the expected CRC value, which is located in the code certificate. If the CRC values match, the FSBL jumps to the OEM_BL.

The code certificate must be programmed into MRAM, and its location in MRAM must be programmed into the SACC0n register or SACC1n register. The SACC0n register specifies the start address of the code certificate when BTFLG = 1. The SACC1n register specifies the start address of the code certificate when BTFLG = 0. Because SACC0n and SACC1n registers are OTP and cannot be erased, there are four areas each for updating the code certificate start address. FSBL reads the values in order from the register with the largest n value, and uses the first value that is not all 1 as the start address of the code certificate. Therefore, use registers in order from the smallest n value.

When measurement reporting is enabled in the FSBLCTRL1 register, the FSBL stores the measurement report to the SRAM address specified by the SAMR register. When reading the measurement report, disable the ECC function by setting SRAMCRn.ECCMOD[1:0] = 00b.

FSBL execution can be skipped after a software reset or deep software standby reset, as configured by the FSBLCTRL0 register.

If the generated HMAC or CRC value is different from the expected value, the high level is output to the port set by the FSBLCTRL2 register and the MCU goes to CPU Sleep mode.

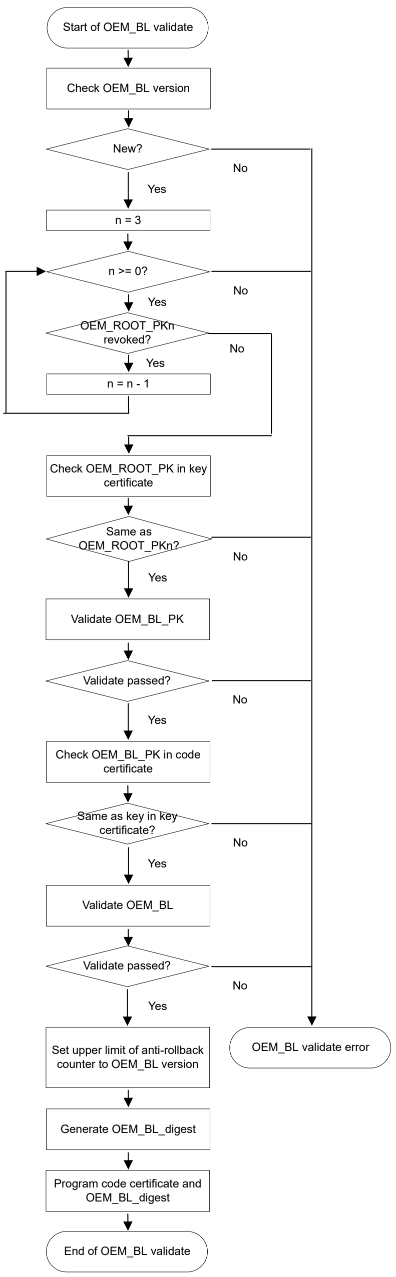


Figure 50.11 OEM_BL validation flow in serial programming mode

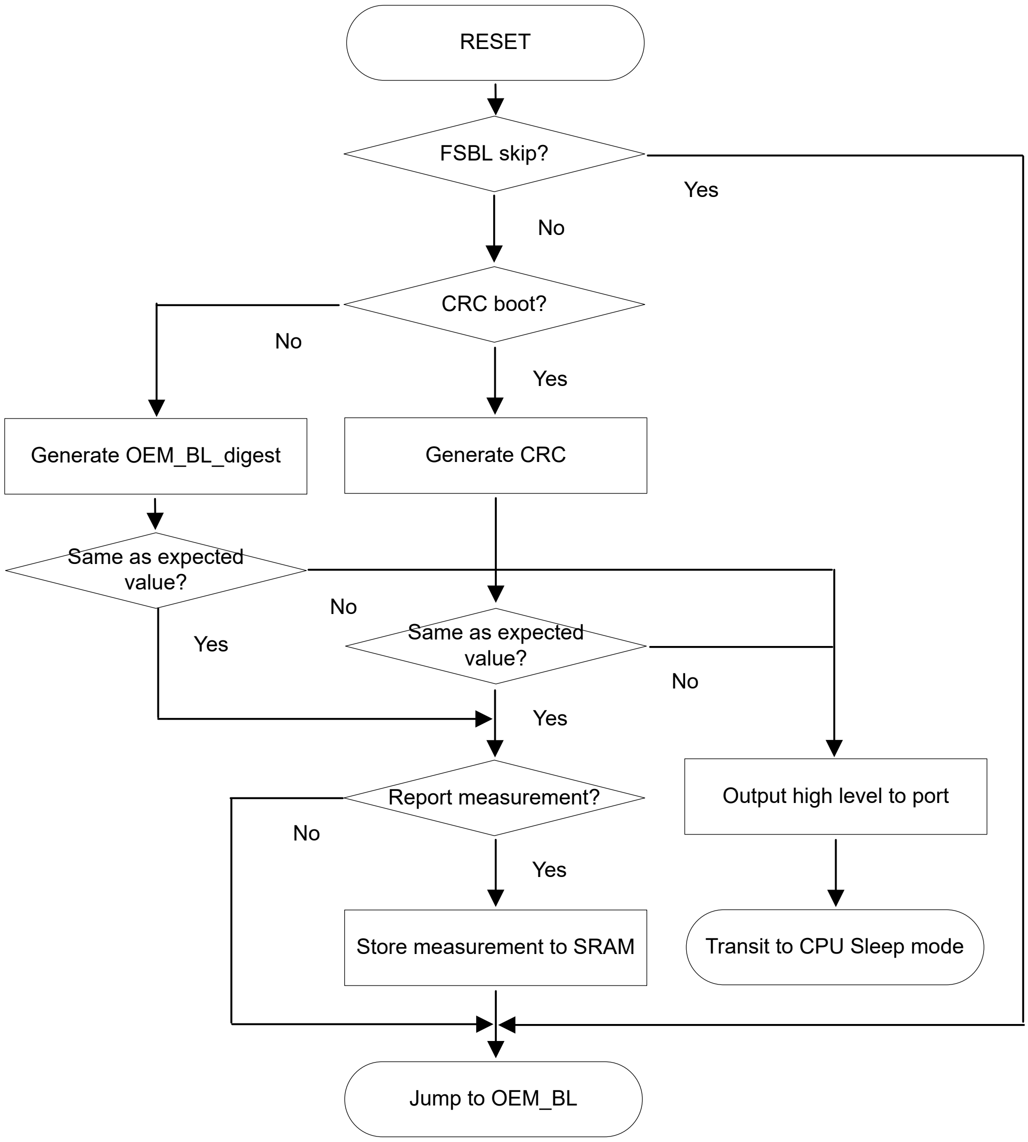


Figure 50.12 FSBL flow in single-chip mode

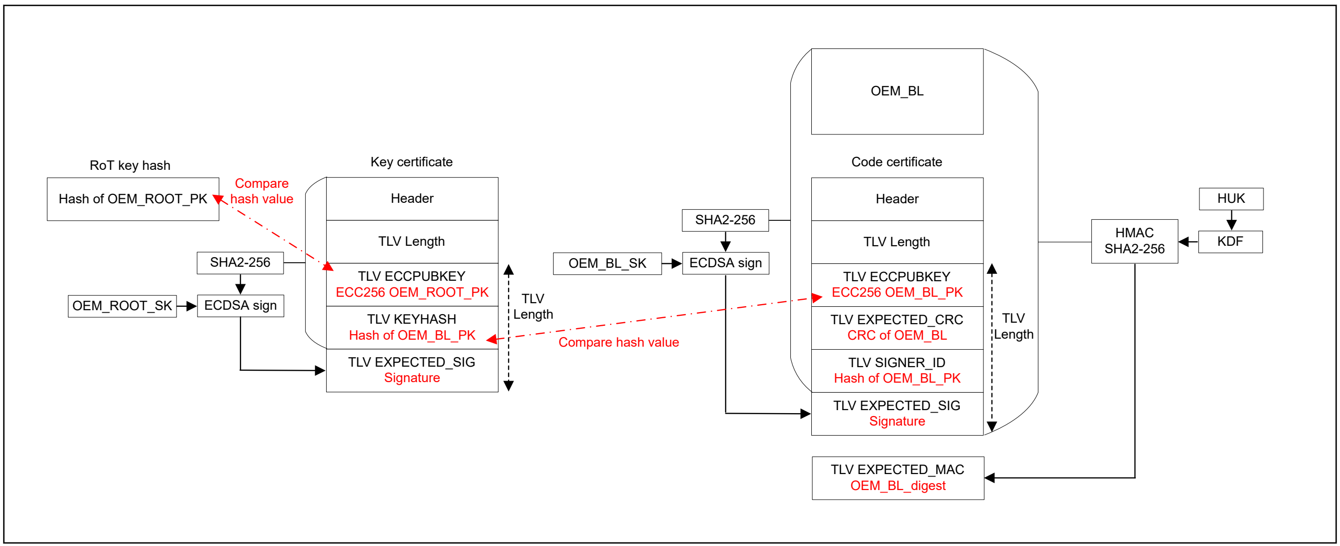


Figure 50.13 Validation structure and certificate format

Table 50.9 Detail of key certificate

Field		Size [byte]	Description
Header	Magic	4	Set to 0x6B657963
	Manifest version	4	Set to 0x00010000
	Flags	4	Reserved for future use
	Reserved	20	Reserved for future use
TLV Length		4	Length in bytes of TLV field
TLV ECCPUBKEY	Type & Length	4	Set to 0x00088010
	Value	64	ECC P-256 OEM_ROOT_PK
TLV KEYHASH	Type & Length	4	Set to 0x10144008
	Value	32	SHA2-256 hash value of OEM_BL_PK
TLV EXPECTED_SIG	Type & Length	4	Set to 0x20088410
	Value	64	Signature

Table 50.10 Detail of code certificate (1 of 2)

Field		Size [byte]	Description
Header	Magic	4	Set to 0x636F6463
	Manifest version	4	Set to 0x00010000
	Flags	4	Set to 0x00000000
	Load Addr	4	Set to 0x00000000
	Dest Addr	4	Set to 0x00000000
	Image size	4	Size in bytes of OEM_BL set in multiples of 16, minimum size is 64 bytes
	Image version	4	Version number of OEM_BL. Can be specified from 1 to 64
	Build number	4	Set to 0x00000000
TLV length		4	Length in bytes of TLV field
TLV ECCPUBKEY	Type & Length	4	Set to 0x01088010
	Value	64	ECC P-256 OEM_BL_PK
TLV EXPECTED_CRC	Type & Length	4	Set to 0x40000001
	Value	4	CRC32 of OEM_BL

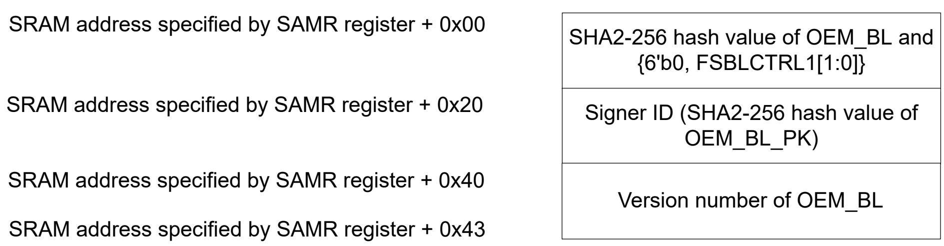
Table 50.10 Detail of code certificate (2 of 2)

Field		Size [byte]	Description
TLV SIGNER_ID	Type & Length	4	Set to 0x10144008
	Value	32	SHA2-256 hash value of OEM_BL_PK
TLV EXPECTED_SIG	Type & Length	4	Set to 0x25088410
	Value	64	Signature of [Code Certificate | OEM_BL], signed by the OEM_BL_SK When generating the signature, enter the code certificate and OEM_BL in the hash function in this order.

Table 50.11 Detail of OEM_BL_digest

Field		Size [byte]	Description
TLV EXPECTED_MAC	Type & Length	4	Fixed value 0x30184008
	Value	32	Unique OEM_BL_digest in each MCU

Note: OEM_BL_digest should be programmed in the area contiguous with the code certificate.

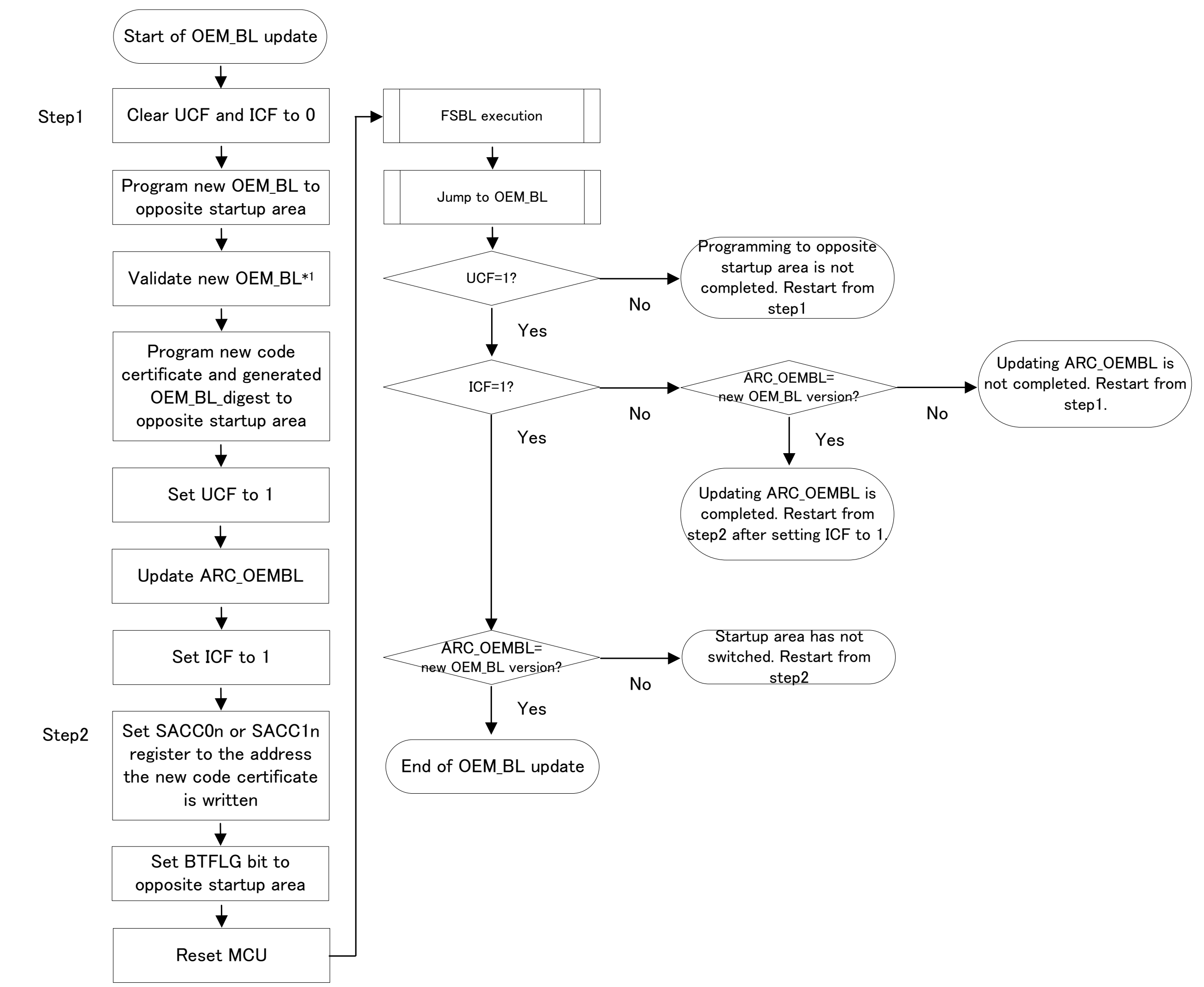


Note: OEM_BL and FSBLCTRL[7:0] are entered into the hash function in this order.

Figure 50.14 Format and location of the measurement report

OEM_BL can be updated by application code. In this case, the application code must validate the updated bootloader, generate the OEM_BL_digest, and store it immediately after the new code certificate, as per the flow described by Figure 50.11.Figure 50.15 show the OEM_BL update flow.

The implementation of the OEM_BL is user defined and application specific. However, it is recommended to implement status flags such as Update Complete Flag (UCF) and Increment Complete Flag (ICF) in the code MRAM as a countermeasure against update errors due to unexpected power failure. If an update error occurs, it can be recovered as shown in the following figures. If an update error occurs, the current OEM_BL is executed after FSBL execution. Note that the update error determination program is required for both the current OEM_BL and the new OEM_BL.



Note 1. Validate new OEM_BL the same way as Figure 50.11.

Figure 50.15 OEM_BL update flow

50.8 Register Description

50.8.1 PSARB : Peripheral Security Attribution Register B

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PSARB31	PSARB30	PSARB29	PSARB28	PSARB27	PSARB26	PSARB25	PSARB24	PSARB23	PSARB22	—	—	PSARB19	PSARB18	PSARB17	PSARB16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	PSARB11	—	PSARB9	PSARB8	PSARB7	—	—	PSARB4	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	PSARB4	I3C Bus Interface Security Attribution Target module: I3C and the MSTPCRB.MSTPB4 bit 0: Secure 1: Non-secure	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	PSARB7	I2C Bus Interface 2 Security Attribution Target module: IIC2 and the MSTPCRB.MSTPB7 bit 0: Secure 1: NonSecure	R/W
8	PSARB8	I2C Bus Interface 1 Security Attribution Target module: IIC1 and the MSTPCRB.MSTPB8 bit 0: Secure 1: Non-secure	R/W
9	PSARB9	I2C Bus Interface 0 Security Attribution Target module: IIC0 and the MSTPCRB.MSTPB9 bit 0: Secure 1: Non-secure	R/W
10	—	This bit is read as 0. The write value should be 0.	R/W
11	PSARB11	Universal Serial Bus 2.0 FS Interface 0 Security Attribution Target module: USBFS0 and the MSTPCRB.MSTPB11 bit 0: Secure 1: Non-secure	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	PSARB16	Octa Memory Controller 0 Security Attribution Target module: OSPI0 (+DOTF0) and the MSTPCRB.MSTPB16 bit 0: Secure 1: Non-secure	R/W
17	PSARB17	Octa Memory Controller 1 Security Attribution Target module: OSPI1 (+DOTF1) and the MSTPCRB.MSTPB17 bit 0: Secure 1: Non-secure	R/W
18	PSARB18	Serial Peripheral Interface 1 Security Attribution Target module: RSPI1 and the MSTPCRB.MSTPB18 bit 0: Secure 1: Non-secure	R/W
19	PSARB19	Serial Peripheral Interface 0 Security Attribution Target module: RSPI0 and the MSTPCRB.MSTPB19 bit 0: Secure 1: Non-secure	R/W
21:20	—	These bits are read as 0. The write value should be 0.	R/W
22	PSARB22	Serial Communication Interface 9 Security Attribution Target module: SCI9 and the MSTPCRB.MSTPB22 bit 0: Secure 1: Non-secure	R/W
23	PSARB23	Serial Communication Interface 8 Security Attribution Target module: SCI8 and the MSTPCRB.MSTPB23 bit 0: Secure 1: Non-secure	R/W
24	PSARB24	Serial Communication Interface 7 Security Attribution Target module: SCI7 and the MSTPCRB.MSTPB24 bit 0: Secure 1: Non-secure	R/W
25	PSARB25	Serial Communication Interface 6 Security Attribution Target module: SCI6 and the MSTPCRB.MSTPB25 bit 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
26	PSARB26	Serial Communication Interface 5 Security Attribution Target module: SCI5 and the MSTPCRB.MSTPB26 bit 0: Secure 1: Non-secure	R/W
27	PSARB27	Serial Communication Interface 4 Security Attribution Target module: SCI4 and the MSTPCRB.MSTPB27 bit 0: Secure 1: Non-secure	R/W
28	PSARB28	Serial Communication Interface 3 Security Attribution Target module: SCI3 and the MSTPCRB.MSTPB28 bit 0: Secure 1: Non-secure	R/W
29	PSARB29	Serial Communication Interface 2 Security Attribution Target module: SCI2 and the MSTPCRB.MSTPB29 bit 0: Secure 1: Non-secure	R/W
30	PSARB30	Serial Communication Interface 1 Security Attribution Target module: SCI1 and the MSTPCRB.MSTPB30 bit 0: Secure 1: Non-secure	R/W
31	PSARB31	Serial Communication Interface 0 Security Attribution Target module: SCI0 and the MSTPCRB.MSTPB31 bit 0: Secure 1: Non-secure	R/W

Note: Access category: S-TYPE-1, P-TYPE-1.

The PSARB register specifies the security attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.2 PSARC : Peripheral Security Attribution Register C

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PSARC31	PSARC30	PSARC29	PSARC28	PSARC27	PSARC26	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	PSARC13	PSARC12	PSARC11	—	—	—	—	—	—	—	—	—	PSARC1	PSARC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PSARC0	Clock Frequency Accuracy Measurement Circuit Security Attribution. Target module: CAC and the MSTPCRC.MSTPC0 bit 0: Secure 1: Non-secure	R/W
1	PSARC1	Cyclic Redundancy Check Calculator Security Attribution Target module: CRC and the MSTPCRC.MSTPC1 bit 0: Secure 1: Non-secure	R/W
10:2	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
11	PSARC11	Secure Digital Host IF 1 Security Attribution Target module: SDHI1 and the MSTPCRC.MSTPC11 bit 0: Secure 1: Non-secure	R/W
12	PSARC12	Secure Digital Host IF 0 Security Attribution Target module: SDHI0 and the MSTPCRC.MSTPC12 bit 0: Secure 1: Non-secure	R/W
13	PSARC13	Data Operation Circuit Security Attribution Target module: DOC and the MSTPCRC.MSTPC13 bit 0: Secure 1: Non-secure	R/W
25:14	—	These bits are read as 0. The write value should be 0.	R/W
26	PSARC26	Controller Area Network with Flexible Data-Rate 1 Security Attribution Target module: CANFD1 and the MSTPCRC.MSTPC26 bit 0: Secure 1: Non-secure	R/W
27	PSARC27	Controller Area Network with Flexible Data-Rate 0 Security Attribution Target module: CANFD0 and the MSTPCRC.MSTPC27 bit 0: Secure 1: Non-secure	R/W
28	PSARC28	Ether-PHY clock Security Attribution Target module: ETHPHYCLK and the MSTPCRC.MSTPC28 bit 0: Secure 1: Non-secure	R/W
29	PSARC29	EtherCAT Slave Controller Security Attribution Target module: ESC and the MSTPCRC.MSTPC29 bit 0: Secure 1: Non-secure	R/W
30	PSARC30	Layer 3 Ethernet Switch Module Security Attribution Target module: ESWM and the MSTPCRC.MSTPC30 bit 0: Secure 1: Non-secure	R/W
31	PSARC31	RSIP-E50D Security Attribution Target module: RSIP-E50D and the MSTPCRC.MSTPC31 bit 0: Secure 1: Non-secure	R/W

Note: Access category: S-TYPE-1, P-TYPE-1.

The PSARC register specifies the security attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.3 PSARD : Peripheral Security Attribution Register D

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x0C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	PSARD28	PSARD27	PSARD26	PSARD25	—	—	PSARD22	PSARD21	PSARD20	PSARD19	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	PSARD14	PSARD13	PSARD12	PSARD11	—	PSARD9	PSARD8	—	PSARD6	PSARD5	PSARD4	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	PSARD4	Asynchronous General Purpose Timer 1 Security Attribution Target module: AGT1 and the MSTPCRD.MSTPD4 bit 0: Secure 1: Non-secure	R/W
5	PSARD5	Asynchronous General Purpose Timer 0 Security Attribution Target module: AGT0 and the MSTPCRD.MSTPD5 bit 0: Secure 1: Non-secure	R/W
6	PSARD6	PWM Delay Generation Circuit Security Attribution Target module: PDG and the MSTPCRD.MSTPD6 bit 0: Secure 1: Non-secure	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	PSARD8	Delta-Sigma Modulator Interface 1 Security Attribution Target module: DSMIF1 and the MSTPCRD.MSTPD8 bit 0: Secure 1: Non-secure	R/W
9	PSARD9	Delta-Sigma Modulator Interface 0 Security Attribution Target module: DSMIF0 and the MSTPCRD.MSTPD9 bit 0: Secure 1: Non-secure	R/W
10	—	This bit is read as 0. The write value should be 0.	R/W
11	PSARD11	Port Output Enable for GPT Group 3 Security Attribution Target module: PGI3 and the MSTPCRD.MSTPD11 bit 0: Secure 1: Non-secure	R/W
12	PSARD12	Port Output Enable for GPT Group 2 Security Attribution Target module: PGI2 and the MSTPCRD.MSTPD12 bit 0: Secure 1: Non-secure	R/W
13	PSARD13	Port Output Enable for GPT Group 1 Security Attribution Target module: PGI1 and the MSTPCRD.MSTPD13 bit 0: Secure 1: Non-secure	R/W
14	PSARD14	Port Output Enable for GPT Group 0 Security Attribution Target module: PGI0 and the MSTPCRD.MSTPD14 bit 0: Secure 1: Non-secure	R/W
18:15	—	These bits are read as 0. The write value should be 0.	R/W
19	PSARD19	12-Bit D/A Converter 1 Security Attribution Target module: DAC121 and the MSTPCRD.MSTPD19 bit 0: Secure 1: Non-secure	R/W
20	PSARD20	12-Bit D/A Converter 0 Security Attribution Target module: DAC120 and the MSTPCRD.MSTPD20 bit 0: Secure 1: Non-secure	R/W
21	PSARD21	16-Bit A/D Converter Security Attribution Target module: ADC16H and the MSTPCRD.MSTPD21 bit 0: Secure 1: Non-secure	R/W
22	PSARD22	Temperature Sensor Security Attribution Target module: TSN and the MSTPCRD.MSTPD22 bit 0: Secure 1: Non-secure	R/W
24:23	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
25	PSARD25	High speed analog Comparator 3 Security Attribution Target module: ACMPHS3 and the MSTPCRD.MSTPD25 bit 0: Secure 1: Non-secure	R/W
26	PSARD26	High speed analog Comparator 2 Security Attribution Target module: ACMPHS2 and the MSTPCRD.MSTPD26 bit 0: Secure 1: Non-secure	R/W
27	PSARD27	High Speed analog Comparator 1 Security Attribution Target module: ACMPHS1 and the MSTPCRD.MSTPD27 bit 0: Secure 1: Non-secure	R/W
28	PSARD28	High Speed Analog Comparator 0 Security Attribution Target module: ACMPHS0 and the MSTPCRD.MSTPD28 bit 0: Secure 1: Non-secure	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: Access category: S-TYPE-1, P-TYPE-1.

The PSARD register specifies the security attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.4 PSARE : Peripheral Security Attribution Register E

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PSARE31	PSARE30	PSARE29	PSARE28	PSARE27	—	—	—	—	—	PSARE21	PSARE20	PSARE19	PSARE18	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PSARE15	—	—	—	—	—	PSARE9	PSARE8	—	—	—	—	PSARE3	PSARE2	PSARE1	PSARE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PSARE0	WDT1 Security Attribution Target module: WDT1 0: Secure 1: Non-secure	R/W
1	PSARE1	WDT0 Security Attribution Target module: WDT0 0: Secure 1: Non-secure	R/W
2	PSARE2	IWDT Security Attribution Target module: IWDT 0: Secure 1: Non-secure	R/W
3	PSARE3	Real Time Clock Security Attribution Target module: RTC 0: Secure 1: Non-secure	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
8	PSARE8	ULPT1 Security Attribution Target module: ULPT1 and the MSTPCRE.MSTPE8 bit 0: Secure 1: Non-secure	R/W
9	PSARE9	ULPT0 Security Attribution Target module: ULPT0 and the MSTPCRE.MSTPE9 bit 0: Secure 1: Non-secure	R/W
14:10	—	These bits are read as 0. The write value should be 0.	R/W
15	PSARE15	GPT common Security Attribution Target module: GPT GTCLKCR register 0: Secure 1: Non-secure	R/W
17:16	—	These bits are read as 0. The write value should be 0.	R/W
18	PSARE18	General PWM Timer Channel 13 Security Attribution Target module: GPT13 and the MSTPCRE.MSTPE18 bit 0: Secure 1: Non-secure	R/W
19	PSARE19	General PWM Timer Channel 12 Security Attribution Target module: GPT12 and the MSTPCRE.MSTPE19 bit 0: Secure 1: Non-secure	R/W
20	PSARE20	General PWM Timer Channel 11 Security Attribution Target module: GPT11 and the MSTPCRE.MSTPE20 bit 0: Secure 1: Non-secure	R/W
21	PSARE21	General PWM Timer Channel 10 Security Attribution Target module: GPT10 and the MSTPCRE.MSTPE21 bit 0: Secure 1: Non-secure	R/W
26:22	—	These bits are read as 0. The write value should be 0.	R/W
27	PSARE27	General PWM Timer Channel 4 Security Attribution Target module: GPT4 and the MSTPCRE.MSTPE27 bit 0: Secure 1: Non-secure	R/W
28	PSARE28	General PWM Timer Channel 3 Security Attribution Target module: GPT3 and the MSTPCRE.MSTPE28 bit 0: Secure 1: Non-secure	R/W
29	PSARE29	General PWM Timer Channel 2 Security Attribution Target module: GPT2 and the MSTPCRE.MSTPE29 bit 0: Secure 1: Non-secure	R/W
30	PSARE30	General PWM Timer Channel 1 Security Attribution Target module: GPT1 and the MSTPCRE.MSTPE30 bit 0: Secure 1: Non-secure	R/W
31	PSARE31	General PWM Timer Channel 0 Security Attribution Target module: GPT0, GPT_OPS and the MSTPCRE.MSTPE31 bit 0: Secure 1: Non-secure	R/W

Note: Access category: S-TYPE-1, P-TYPE-1.

The PSARE register specifies the security attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.5 MSSAR : Module Stop Security Attribution Register

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x14

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSSAR31	—	—	—	—	—	—	—	MSSAR23	MSSAR22	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	MSSAR3	MSSAR2	MSSAR1	MSSAR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MSSAR0	SRAM0 Clock Stop Security Attribution Target module: the MSTPCRA.MSTPA0 bit 0: Secure 1: Non-secure	R/W
1	MSSAR1	SRAM1 Clock Stop Security Attribution Target module: the MSTPCRA.MSTPA1 bit 0: Secure 1: Non-secure	R/W
2	MSSAR2	SRAM2 Clock Stop Security Attribution Target module: the MSTPCRA.MSTPA2 bit 0: Secure 1: Non-secure	R/W
3	MSSAR3	SRAM3 Clock Stop Security Attribution Target module: the MSTPCRA.MSTPA3 bit 0: Secure 1: Non-secure	R/W
21:4	—	These bits are read as 0. The write value should be 0.	R/W
22	MSSAR22	DMAC0/DTC0 Clock Stop Security Attribution Target module: the MSTPCRA.MSTPA22 bit 0: Secure 1: Non-secure	R/W
23	MSSAR23	DMAC1/DTC1 Clock Stop Security Attribution Target module: the MSTPCRA.MSTPA23 bit 0: Secure 1: Non-secure	R/W
30:24	—	These bits are read as 0. The write value should be 0.	R/W
31	MSSAR31	ELC Clock Stop Security Attribution Target module: the MSTPCRC.MSTPC14 bit 0: Secure 1: Non-secure	R/W

Note: Access category: S-TYPE-1, P-TYPE-1.

The MSSAR register specifies the security attribution for the corresponding bit in Module Stop Control Register.

50.8.6 PPARB : Peripheral Privilege Attribution Register B

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x1C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PPARB31	PPARB30	PPARB29	PPARB28	PPARB27	PPARB26	PPARB25	PPARB24	PPARB23	PPARB22	—	—	PPARB19	PPARB18	PPARB17	PPARB16
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	PPARB11	—	PPARB9	PPARB8	PPARB7	—	—	PPARB4	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 1. The write value should be 1.	R/W
4	PPARB4	I3C Bus Interface Privilege Attribution Target module: I3C and the MSTPCRB.MSTPB4 bit 0: Privileged 1: Unprivileged	R/W
6:5	—	These bits are read as 1. The write value should be 1.	R/W
7	PPARB7	I2C Bus Interface 2 Privilege Attribution Target module: IIC2 and the MSTPCRB.MSTPB7 bit 0: Privileged 1: Unprivileged	R/W
8	PPARB8	I2C Bus Interface 1 Privilege Attribution Target module: IIC1 and the MSTPCRB.MSTPB8 bit 0: Privileged 1: Unprivileged	R/W
9	PPARB9	I2C Bus Interface 0 Privilege Attribution Target module: IIC0 and the MSTPCRB.MSTPB9 bit 0: Privileged 1: Unprivileged	R/W
10	—	This bit is read as 1. The write value should be 1.	R/W
11	PPARB11	Universal Serial Bus 2.0 FS Interface 0 Privilege Attribution Target module: USBFS0 and the MSTPCRB.MSTPB11 bit 0: Privileged 1: Unprivileged	R/W
15:12	—	These bits are read as 1. The write value should be 1.	R/W
16	PPARB16	Octa Memory Controller 0 Privilege Attribution Target module: OSPI0 (+DOTF0) and the MSTPCRB.MSTPB16 bit 0: Privileged 1: Unprivileged	R/W
17	PPARB17	Octa Memory Controller 1 Privilege Attribution Target module: OSPI1 (+DOTF1) and the MSTPCRB.MSTPB17 bit 0: Privileged 1: Unprivileged	R/W
18	PPARB18	Serial Peripheral Interface 1 Privilege Attribution Target module: RSPI1 and the MSTPCRB.MSTPB18 bit 0: Privileged 1: Unprivileged	R/W
19	PPARB19	Serial Peripheral Interface 0 Privilege Attribution Target module: RSPI0 and the MSTPCRB.MSTPB19 bit 0: Privileged 1: Unprivileged	R/W
21:20	—	These bits are read as 1. The write value should be 1.	R/W

Bit	Symbol	Function	R/W
22	PPARB22	Serial Communication Interface 9 Privilege Attribution Target module: SCI9 and the MSTPCRB.MSTPB22 bit 0: Privileged 1: Unprivileged	R/W
23	PPARB23	Serial Communication Interface 8 Privilege Attribution Target module: SCI8 and the MSTPCRB.MSTPB23 bit 0: Privileged 1: Unprivileged	R/W
24	PPARB24	Serial Communication Interface 7 Privilege Attribution Target module: SCI7 and the MSTPCRB.MSTPB24 bit 0: Privileged 1: Unprivileged	R/W
25	PPARB25	Serial Communication Interface 6 Privilege Attribution Target module: SCI6 and the MSTPCRB.MSTPB25 bit 0: Privileged 1: Unprivileged	R/W
26	PPARB26	Serial Communication Interface 5 Privilege Attribution Target module: SCI5 and the MSTPCRB.MSTPB26 bit 0: Privileged 1: Unprivileged	R/W
27	PPARB27	Serial Communication Interface 4 Privilege Attribution Target module: SCI4 and the MSTPCRB.MSTPB27 bit 0: Privileged 1: Unprivileged	R/W
28	PPARB28	Serial Communication Interface 3 Privilege Attribution Target module: SCI3 and the MSTPCRB.MSTPB28 bit 0: Privileged 1: Unprivileged	R/W
29	PPARB29	Serial Communication Interface 2 Privilege Attribution Target module: SCI2 and the MSTPCRB.MSTPB29 bit 0: Privileged 1: Unprivileged	R/W
30	PPARB30	Serial Communication Interface 1 Privilege Attribution Target module: SCI1 and the MSTPCRB.MSTPB30 bit 0: Privileged 1: Unprivileged	R/W
31	PPARB31	Serial Communication Interface 0 Privilege Attribution Target module: SCI0 and the MSTPCRB.MSTPB31 bit 0: Privileged 1: Unprivileged	R/W

Note: Access category: S-TYPE-2, P-TYPE-1.

The PPARB register specifies the privilege attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.7 PPARC : Peripheral Privilege Attribution Register C

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x20

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PPAR C31	PPAR C30	PPAR C29	PPAR C28	PPAR C27	PPAR C26	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	PPAR C13	PPAR C12	PPAR C11	—	—	—	—	—	—	—	—	—	PPAR C1	PPAR C0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	PPARC0	Clock Frequency Accuracy Measurement Circuit Privilege Attribution Target module: CAC and the MSTPCRC.MSTPC0 bit 0: Privileged 1: Unprivileged	R/W
1	PPARC1	Cyclic Redundancy Check Calculator Privilege Attribution Target module: CRC and the MSTPCRC.MSTPC1 bit 0: Privileged 1: Unprivileged	R/W
10:2	—	These bits are read as 1. The write value should be 1.	R/W
11	PPARC11	Secure Digital Host IF 1 Privilege Attribution Target module: SDHI1 and the MSTPCRC.MSTPC11 bit 0: Privileged 1: Unprivileged	R/W
12	PPARC12	Secure Digital Host IF 0 Privilege Attribution Target module: SDHI0 and the MSTPCRC.MSTPC12 bit 0: Privileged 1: Unprivileged	R/W
13	PPARC13	Data Operation Circuit Privilege Attribution Target module: DOC and the MSTPCRC.MSTPC13 bit 0: Privileged 1: Unprivileged	R/W
25:14	—	These bits are read as 1. The write value should be 1.	R/W
26	PPARC26	Controller Area Network with Flexible Data-Rate 1 Privilege Attribution Target module: CANFD1 and the MSTPCRC.MSTPC26 bit 0: Privileged 1: Unprivileged	R/W
27	PPARC27	Controller Area Network with Flexible Data-Rate 0 Privilege Attribution Target module: CANFD0 and the MSTPCRC.MSTPC27 bit 0: Privileged 1: Unprivileged	R/W
28	PPARC28	Ether-PHY clock Privilege Attribution Target module: ETHPHYCLK and the MSTPCRC.MSTPC28 bit 0: Privileged 1: Unprivileged	R/W
29	PPARC29	EtherCAT Slave Controller Privilege Attribution Target module: ESC and the MSTPCRC.MSTPC29 bit 0: Privileged 1: Unprivileged	R/W
30	PPARC30	Layer 3 Ethernet Switch Module Privilege Attribution Target module: ESWM and the MSTPCRC.MSTPC30 bit 0: Privileged 1: Unprivileged	R/W

Bit	Symbol	Function	R/W
31	PPARC31	RSIP-E50D Privilege Attribution Target module: RSIP-E50D and the MSTPCRC.MSTPC31 bit 0: Privileged 1: Unprivileged	R/W

Note: Access category: S-TYPE-2, P-TYPE-1.

The PPARC register specifies the privilege attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.8 PPARD : Peripheral Privilege Attribution Register D

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x24

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	PPARD28	PPARD27	PPARD26	PPARD25	—	—	PPARD22	PPARD21	PPARD20	PPARD19	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	PPARD14	PPARD13	PPARD12	PPARD11	—	PPARD9	PPARD8	—	PPARD6	PPARD5	PPARD4	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 1. The write value should be 1.	R/W
4	PPARD4	Asynchronous General Purpose Timer 1 Privilege Attribution Target module: AGT1 and the MSTPCRD.MSTPD4 bit 0: Privileged 1: Unprivileged	R/W
5	PPARD5	Asynchronous General Purpose Timer 0 Privilege Attribution Target module: AGT0 and the MSTPCRD.MSTPD5 bit 0: Privileged 1: Unprivileged	R/W
6	PPARD6	PWM Delay Generation Circuit Privilege Attribution Target module: PDG and the MSTPCRD.MSTPD6 bit 0: Privileged 1: Unprivileged	R/W
7	—	This bit is read as 1. The write value should be 1.	R/W
8	PPARD8	Delta-Sigma Modulator Interface 1 Privilege Attribution Target module: DSMIF1 and the MSTPCRD.MSTPD8 bit 0: Privileged 1: Unprivileged	R/W
9	PPARD9	Delta-Sigma Modulator Interface 0 Privilege Attribution Target module: DSMIF0 and the MSTPCRD.MSTPD9 bit 0: Privileged 1: Unprivileged	R/W
10	—	This bit is read as 1. The write value should be 1.	R/W
11	PPARD11	Port Output Enable for GPT Group 3 Privilege Attribution Target module: PGI3 and the MSTPCRD.MSTPD11 bit 0: Privileged 1: Unprivileged	R/W
12	PPARD12	Port Output Enable for GPT Group 2 Privilege Attribution Target module: PGI2 and the MSTPCRD.MSTPD12 bit 0: Privileged 1: Unprivileged	R/W

Bit	Symbol	Function	R/W
13	PPARD13	Port Output Enable for GPT Group 1 Privilege Attribution Target module: PGI1 and the MSTPCRD.MSTPD13 bit 0: Privileged 1: Unprivileged	R/W
14	PPARD14	Port Output Enable for GPT Group 0 Privilege Attribution Target module: PGI0 and the MSTPCRD.MSTPD14 bit 0: Privileged 1: Unprivileged	R/W
18:15	—	These bits are read as 1. The write value should be 1.	R/W
19	PPARD19	12-Bit D/A Converter 1 Privilege Attribution Target module: DAC121 and the MSTPCRD.MSTPD19 bit 0: Privileged 1: Unprivileged	R/W
20	PPARD20	12-Bit D/A Converter 0 Privilege Attribution Target module: DAC120 and the MSTPCRD.MSTPD20 bit 0: Privileged 1: Unprivileged	R/W
21	PPARD21	16-Bit A/D Converter Privilege Attribution Target module: ADC16H and the MSTPCRD.MSTPD21 bit 0: Privileged 1: Unprivileged	R/W
22	PPARD22	Temperature Sensor Privilege Attribution Target module: TSN and the MSTPCRD.MSTPD22 bit 0: Privileged 1: Unprivileged	R/W
24:23	—	These bits are read as 1. The write value should be 1.	R/W
25	PPARD25	High speed analog Comparator 3 Privilege Attribution Target module: ACMPHS3 and the MSTPCRD.MSTPD25 bit 0: Privileged 1: Unprivileged	R/W
26	PPARD26	High speed analog Comparator 2 Privilege Attribution Target module: ACMPHS2 and the MSTPCRD.MSTPD26 bit 0: Privileged 1: Unprivileged	R/W
27	PPARD27	High speed analog Comparator 1 Privilege Attribution Target module: ACMPHS1 and the MSTPCRD.MSTPD27 bit 0: Privileged 1: Unprivileged	R/W
28	PPARD28	High speed analog Comparator 0 Privilege Attribution Target module: ACMPHS0 and the MSTPCRD.MSTPD28 bit 0: Privileged 1: Unprivileged	R/W
31:29	—	These bits are read as 1. The write value should be 1.	R/W

Note: Access category: S-TYPE-2, P-TYPE-1.

The PPARD register specifies the privilege attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.9 PPARE : Peripheral Privilege Attribution Register E

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x28

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PPARE31	PPARE30	PPARE29	PPARE28	PPARE27	—	—	—	—	—	PPARE21	PPARE20	PPARE19	PPARE18	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PPARE15	—	—	—	—	—	PPARE9	PPARE8	—	—	—	—	PPARE3	PPARE2	PPARE1	PPARE0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	PPARE0	WDT1 Privilege Attribution Target module: WDT1 0: Privileged 1: Unprivileged	R/W
1	PPARE1	WDT0 Privilege Attribution Target module: WDT0 0: Privileged 1: Unprivileged	R/W
2	PPARE2	IWDT Privilege Attribution Target module: IWDT 0: Privileged 1: Unprivileged	R/W
3	PPARE3	Real Time Clock Privilege Attribution Target module: RTC 0: Privileged 1: Unprivileged	R/W
7:4	—	These bits are read as 1. The write value should be 1.	R/W
8	PPARE8	ULPT1 Privilege Attribution Target module: ULPT1 and the MSTPCRE.MSTPE8 bit 0: Privileged 1: Unprivileged	R/W
9	PPARE9	ULPT0 Privilege Attribution Target module: ULPT0 and the MSTPCRE.MSTPE9 bit 0: Privileged 1: Unprivileged	R/W
14:10	—	These bits are read as 1. The write value should be 1.	R/W
15	PPARE15	GPT common Privilege Attribution Target module: GPT GTCLKCR register 0: Privileged 1: Unprivileged	R/W
17:16	—	These bits are read as 1. The write value should be 1.	R/W
18	PPARE18	General PWM Timer Channel 13 Privilege Attribution Target module: GPT13 and the MSTPCRE.MSTPE18 bit 0: Privileged 1: Unprivileged	R/W
19	PPARE19	General PWM Timer Channel 12 Privilege Attribution Target module: GPT12 and the MSTPCRE.MSTPE19 bit 0: Privileged 1: Unprivileged	R/W

Bit	Symbol	Function	R/W
20	PPARE20	General PWM Timer Channel 11 Privilege Attribution Target module: GPT11 and the MSTPCRE.MSTPE20 bit 0: Privileged 1: Unprivileged	R/W
21	PPARE21	General PWM Timer Channel 10 Privilege Attribution Target module: GPT10 and the MSTPCRE.MSTPE21 bit 0: Privileged 1: Unprivileged	R/W
26:22	—	These bits are read as 1. The write value should be 1.	R/W
27	PPARE27	General PWM Timer Channel 4 Privilege Attribution Target module: GPT4 and the MSTPCRE.MSTPE27 bit 0: Privileged 1: Unprivileged	R/W
28	PPARE28	General PWM Timer Channel 3 Privilege Attribution Target module: GPT3 and the MSTPCRE.MSTPE28 bit 0: Privileged 1: Unprivileged	R/W
29	PPARE29	General PWM Timer Channel 2 Privilege Attribution Target module: GPT2 and the MSTPCRE.MSTPE29 bit 0: Privileged 1: Unprivileged	R/W
30	PPARE30	General PWM Timer Channel 1 Privilege Attribution Target module: GPT1 and the MSTPCRE.MSTPE30 bit 0: Privileged 1: Unprivileged	R/W
31	PPARE31	General PWM Timer Channel 0 Privilege Attribution Target module: GPT0 and the MSTPCRE.MSTPE31 bit 0: Privileged 1: Unprivileged	R/W

Note: Access category: S-TYPE-2, P-TYPE-1.

The PPARE register specifies the privilege attribution for each module and the corresponding bit in Module Stop Control Register.

50.8.10 MSPAR : Module Stop Privilege Attribution Register

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x2C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MSPAR31	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
30:0	—	These bits are read as 1. The write value should be 1.	R/W
31	MSPAR31	ELC Clock Stop Privilege Attribution Target module: the MSTPCRC.MSTPC14 bit 0: Privileged 1: Unprivileged	R/W

Note: Access category: S-TYPE-2, P-TYPE-1.

The MSPAR register specifies the privilege attribution for the corresponding bit in the Module Stop Control Register.

50.8.11 CMSAMON : Code MRAM Security Attribution Monitor Register

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x30

Bit position:	31								23								15															0
Bit field:	—	—	—	—	—	—	—	—	CMS[8:0]									—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	*1	*1	*1	*1	*1	*1	*1	*1	*1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	—	These bits are read as 0.	R
23:15	CMS[8:0]	Code MRAM Secure Area Indicate the area of secure region for code MRAM.	R
31:24	—	These bits are read as 0.	R

Note: Access category: S-TYPE-5, P-TYPE-5.
Note 1. The value in a blank product is 0x1FF. It is set to the value written by your application.

50.8.12 SFSAMON : SiP Flash Security Attribution Monitor Register

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x3C

Bit position:	31								23								15															0
Bit field:	—	—	—	—	—	—	—	—	SFS[8:0]									—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	*1	*1	*1	*1	*1	*1	*1	*1	*1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	—	These bits are read as 0.	R
23:15	SFS[8:0]	SiP Flash Secure Area Indicate the area of secure region for SiP flash.	R
31:24	—	These bits are read as 0.	R

Note: Access category: S-TYPE-5, P-TYPE-5.
Note 1. The value in a blank product is 0x1FF. It is set to the value written by your application.

50.8.13 DLMMON : Device Lifecycle Management State Monitor Register

Base address: PSCU = 0x4020_4000
PSCU_NS = 0x5020_4000

Offset address: 0x38

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	DLMMON[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	*1	*1	*1	*1

Bit	Symbol	Function	R/W
3:0	DLMMON[3:0]	Device Lifecycle Management State Monitor Indicate DLM status value. 0x4: OEM 0x6: LCK_BOOT 0x7: RMA_REQ 0x8: RMA_ACK 0x9: RMA_RET Others: Reserved	R
31:4	—	These bits are read as 0.	R

Note: Access category: S-TYPE-5, P-TYPE-5.
Note 1. The value in a blank product is 0100b. These bits depend on DLM status.

50.8.14 MSAOAD : Master Security Attribution Operation After Detection Register

Base address: BUS = 0x4000_3000

Offset address: 0x1010

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	OAD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OAD	Operation after detection 0: IRQ 1: Reset	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code These bits enable or disable writing of the OAD bit.	W

Note: Access category: S-TYPE-6, P-TYPE-2.

OAD bit (Operation after detection)

The OAD bit specifies operation when the access violation is detected. When OAD = 0, error response is returned to the bus master, and IRQ request is generated if BUSIRQEN<Master Name>. EN bit is 1. Interrupts due to bus errors can be used as NMI if NMIER.BUSEN bit is 1. When OAD = 1, reset request is generated. When writing to the OAD bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the OAD bit. When writing to the OAD bit, write 0xA5 simultaneously to the KEY[7:0] bits. If this value is not written to the KEY[7:0] bits, the OAD bit is not updated.

50.8.15 MSAPT : Master Security Attribution Protect Register

Base address: BUS = 0x4000_3000

Offset address: 0x1014

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MSAOAD register writing is possible. 1: MSAOAD register writing is protected. Read is possible.	R/W

Bit	Symbol	Function	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code These bits enable or disable writing of the PROTECT bit.	W

Note: Access category: S-TYPE-6, P-TYPE-2.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writes to the associated registers to be protected.

MSAPT.PROTECT controls the MSAOAD register.

When the PROTECT bit is set simultaneously, write 0xA5 to the KEY[7:0] bits using half word access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. If this value is not written to the KEY[7:0] bits, the PROTECT bit is not updated.

50.9 Usage Note

50.9.1 Security or Privilege Bit Write Timing

When writing a security or privilege attribution bit, confirm that writing is complete by reading the security or privilege register until it matches the value written. The protection is not effective until writing to the security or privilege register is complete.